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(54) **RESONANT SWITCHED CAPACITOR
CONVERTER CONTROL MODULE AND
METHOD WITH ZERO CURRENT
SWITCHING CONTROL**

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H02M 1/00 (2006.01)

H02M 3/07 (2006.01)

(52) **U.S. Cl.**

CPC **H02M 3/01** (2021.05); **H02M 1/0058**
(2021.05); **H02M 3/07** (2013.01)

(58) **Field of Classification Search**

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See application file for complete search history.

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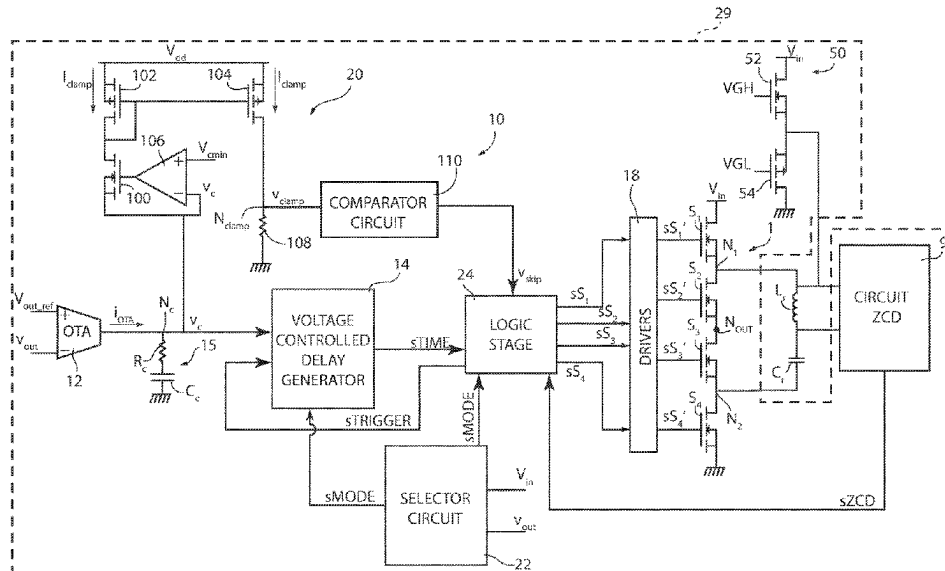
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(57) **ABSTRACT**

A control module, for a resonant switched-capacitor con-
verter with an inductor, includes a controller stage, an input
stage generating a control signal indicating a control quan-
tity, a delay stage generating a duration signal indicating a
time quantity, and a circuit indicating zero crossings of the
inductor current. If the control quantity is variable, the input
stage clamps the control quantity to a control threshold.
When in normal mode, the controller stage controls the
converter to carry out a phase sequence with timings that
depend on the zero crossings and the time quantity, so the
converter generates an output current that depends on the
time quantity and is prevented from dropping below a
minimum current. If the output voltage reaches an upper
threshold, the controller stage switches into pulse-skipping
mode to suspend the phase sequence, and resumes the phase
sequence after the output voltage drops to a lower threshold.

20 Claims, 11 Drawing Sheets



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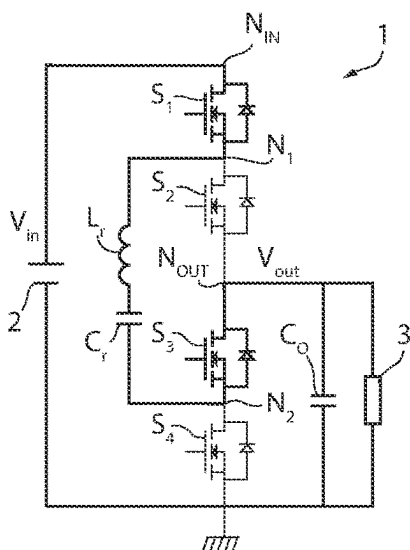


Fig. 1A

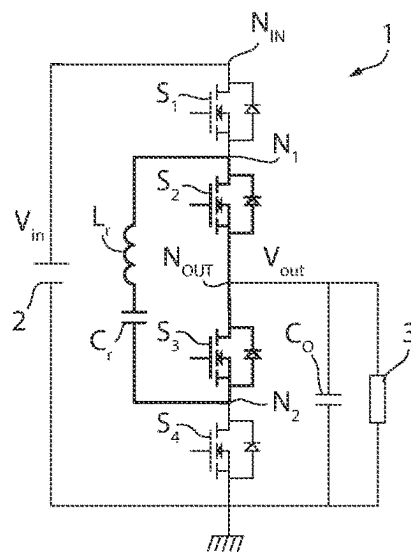


Fig. 1B

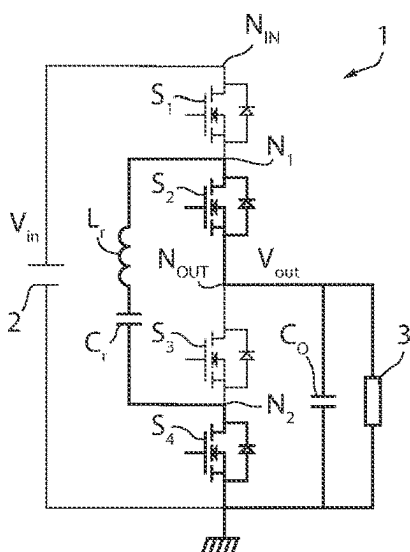


Fig. 1C

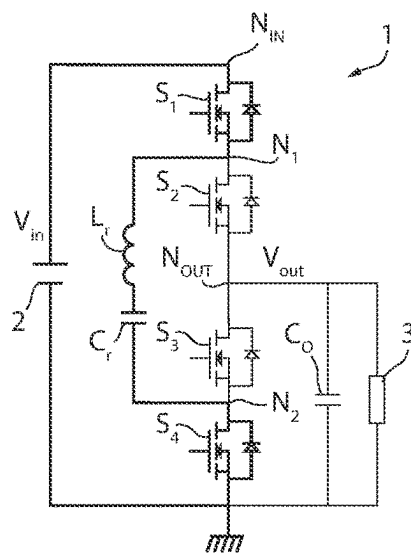


Fig. 1D

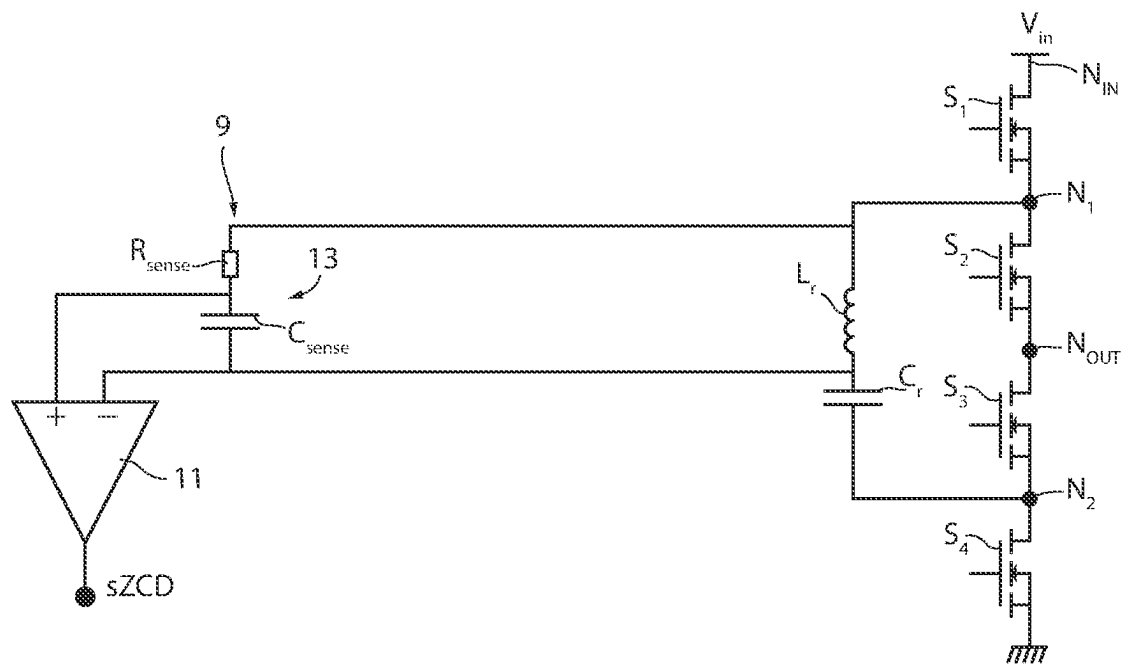


Fig. 2

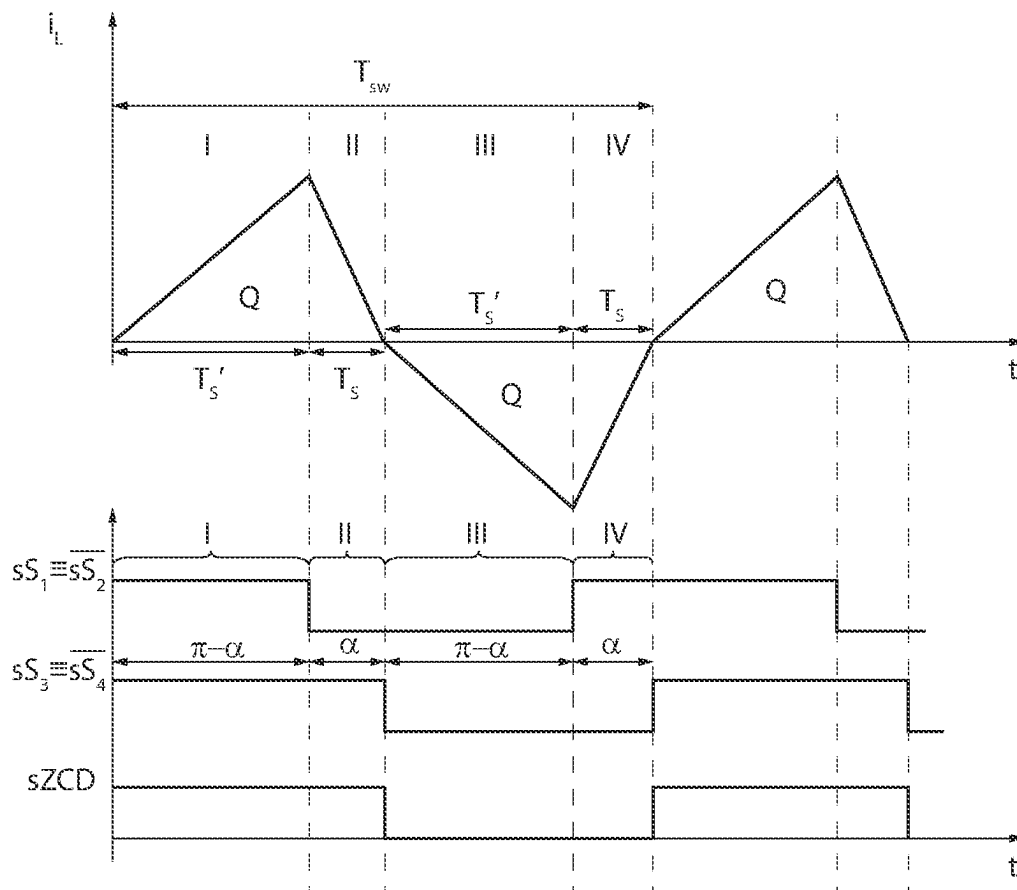


Fig. 3A

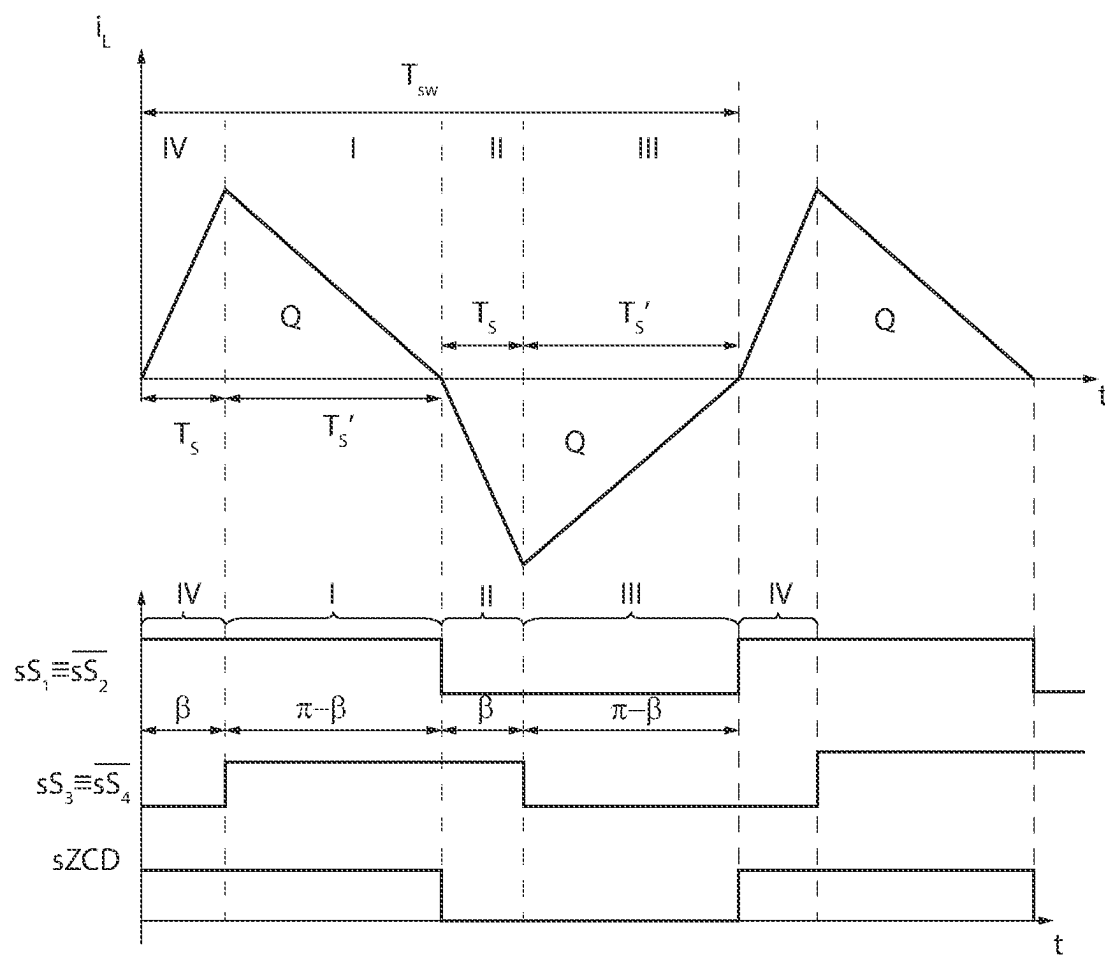


Fig. 3B

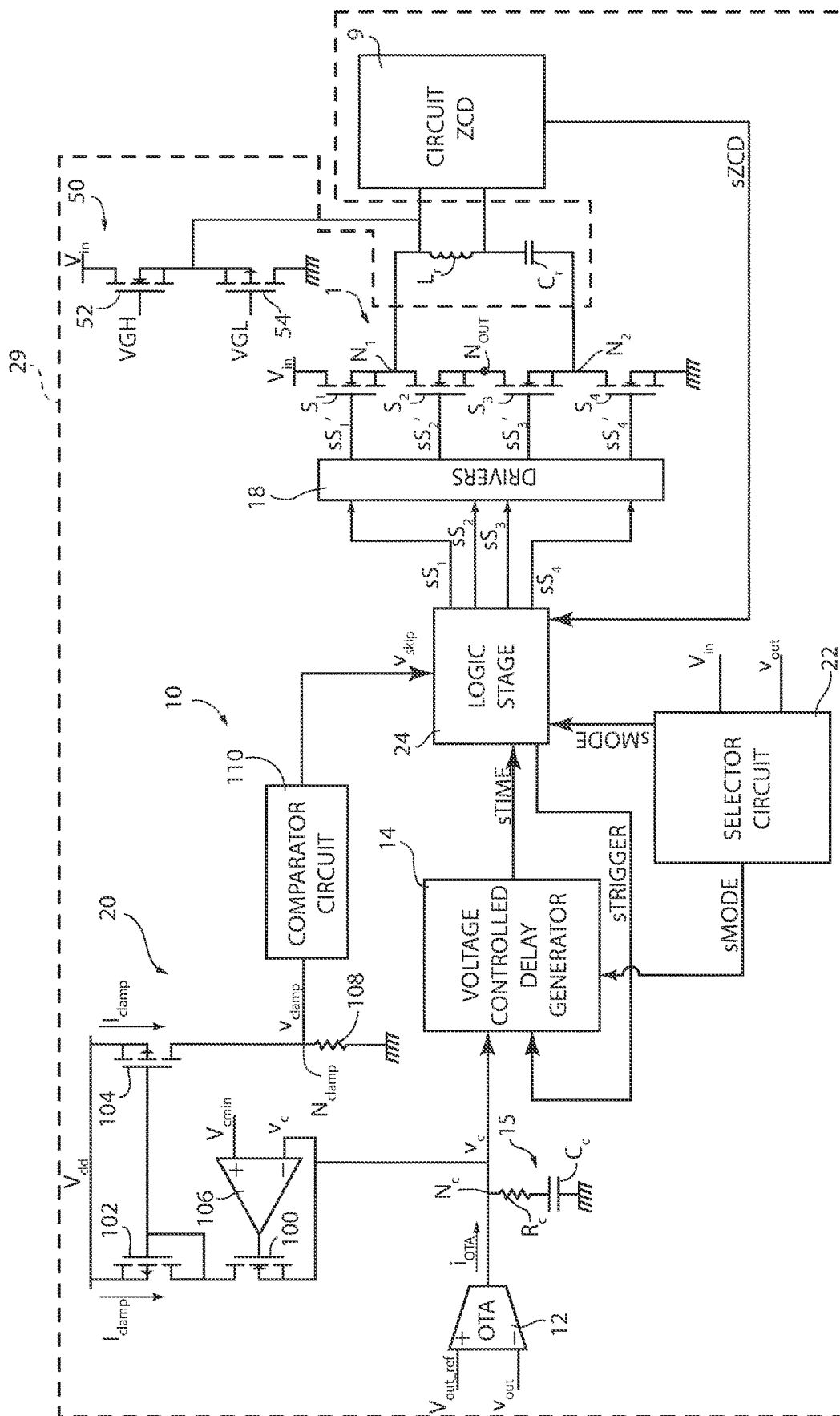


Fig. 4

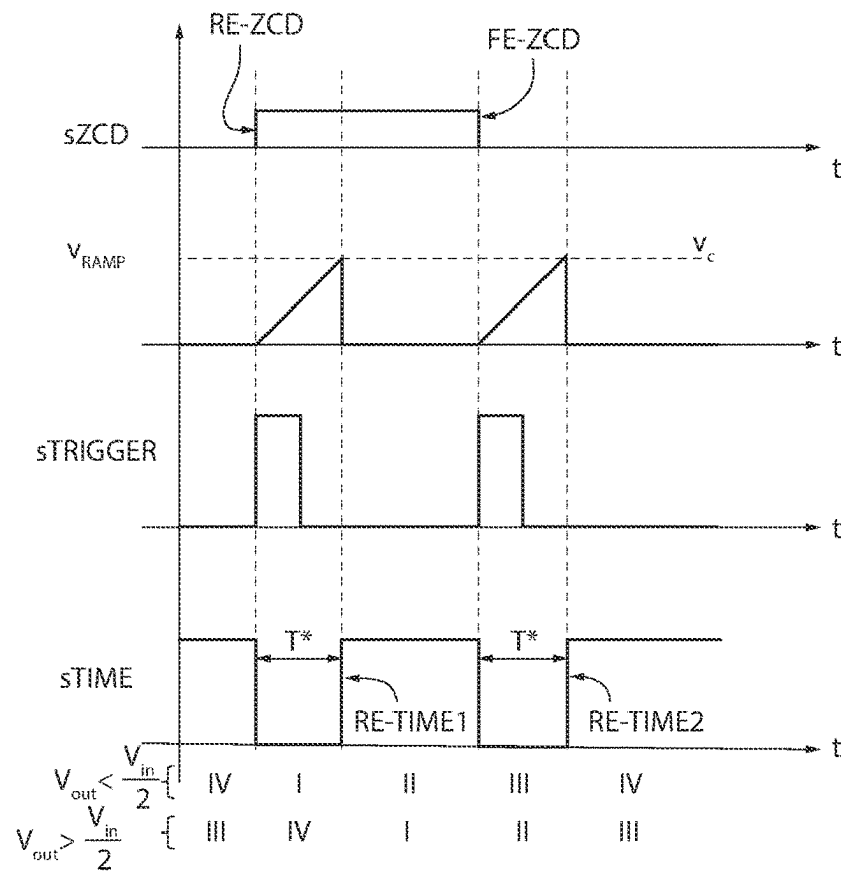


Fig. 5

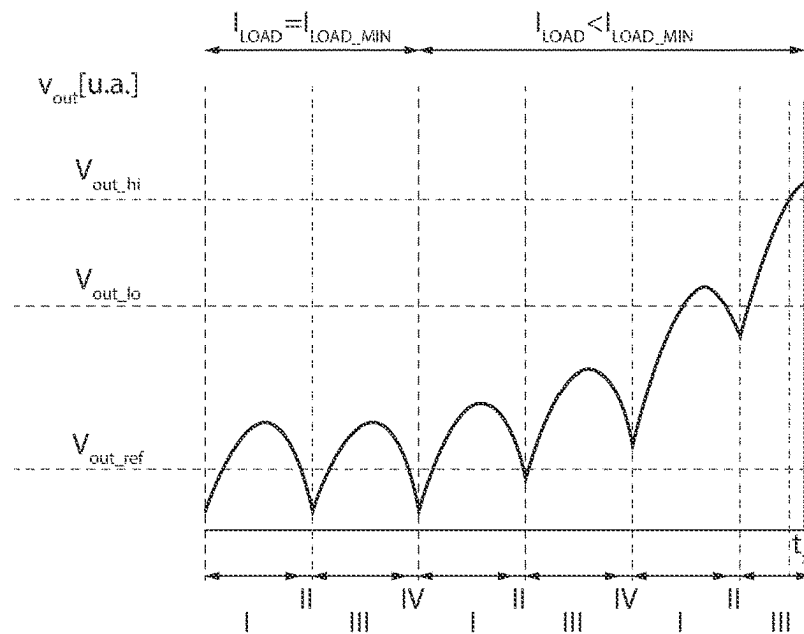


Fig. 6

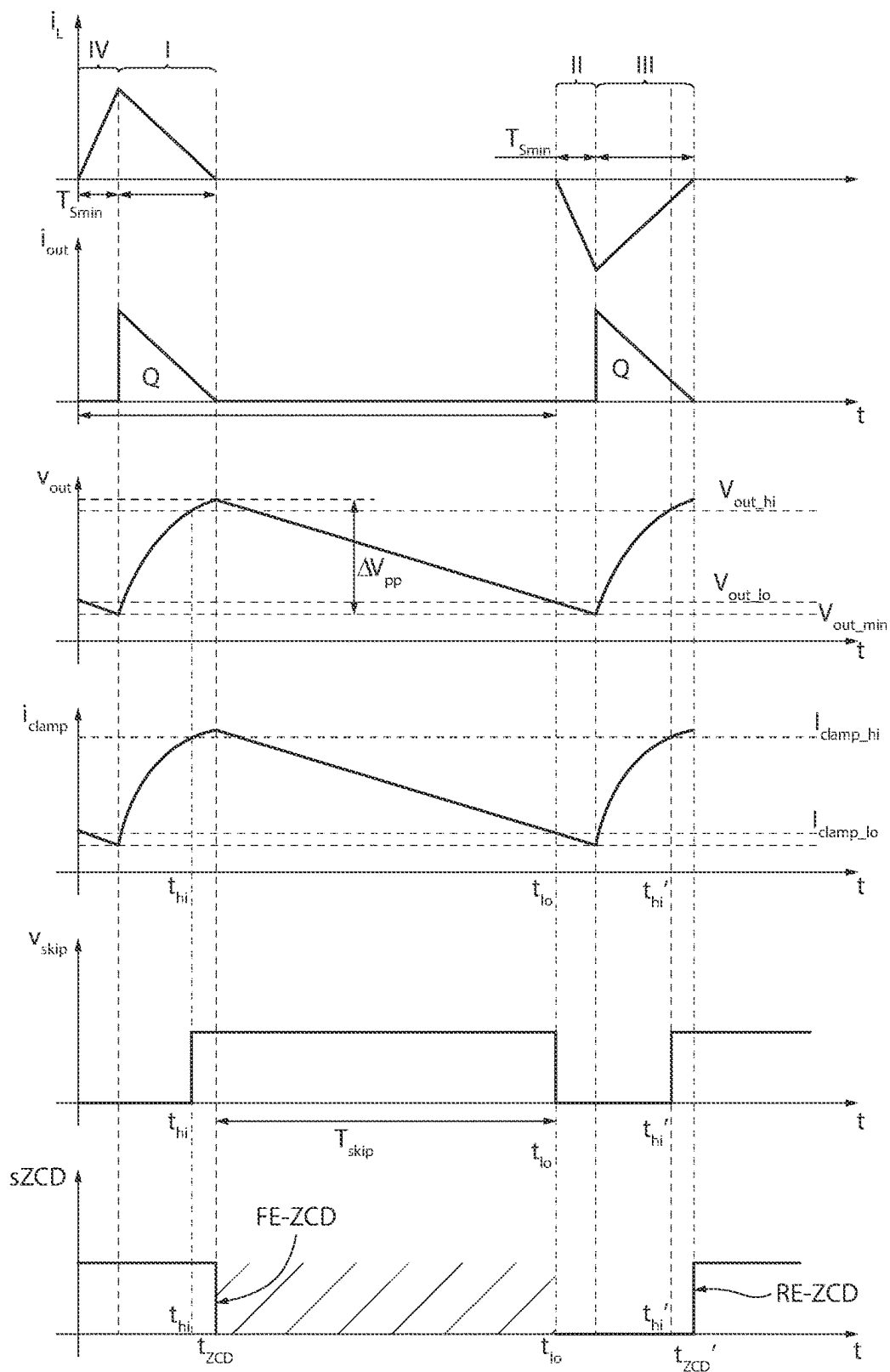


Fig. 7

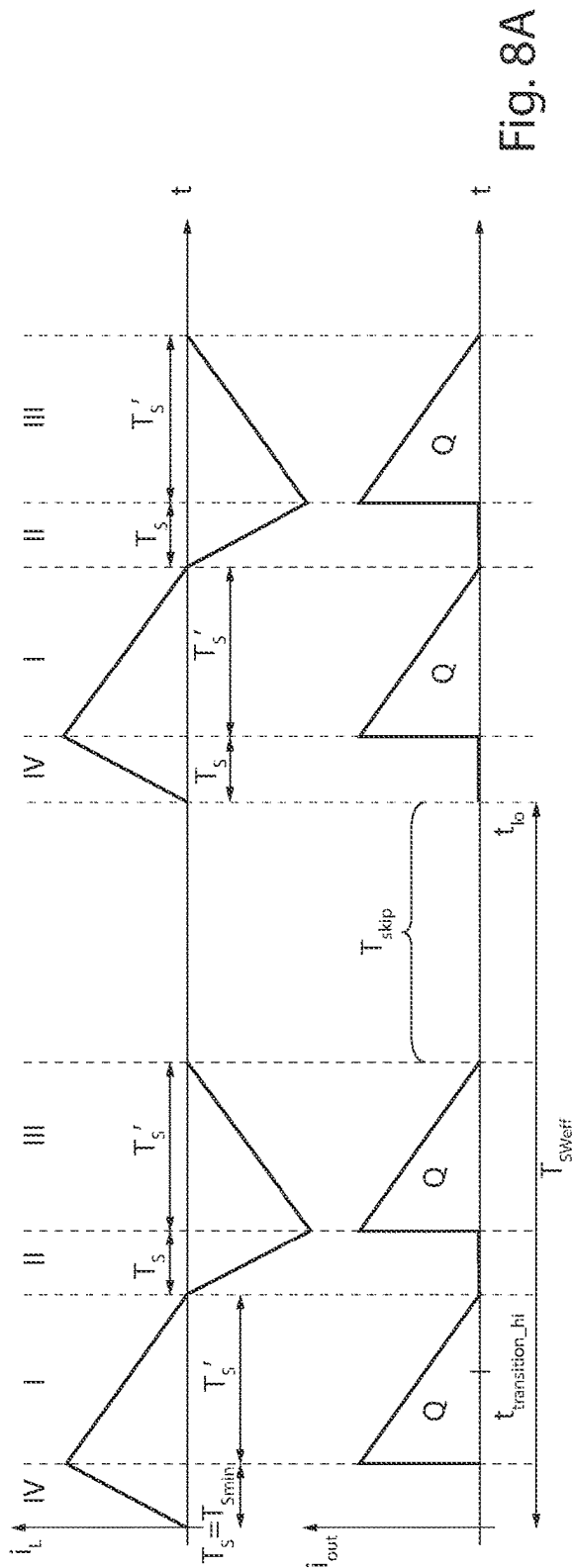


Fig. 8A

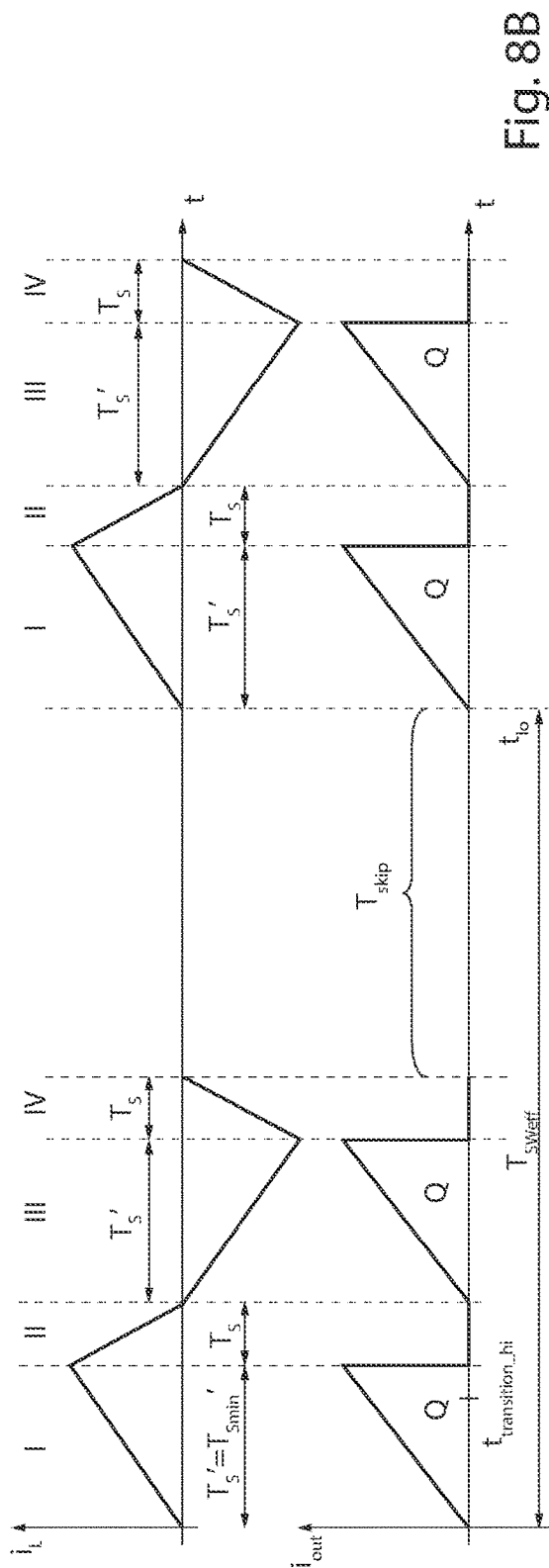


Fig. 8B

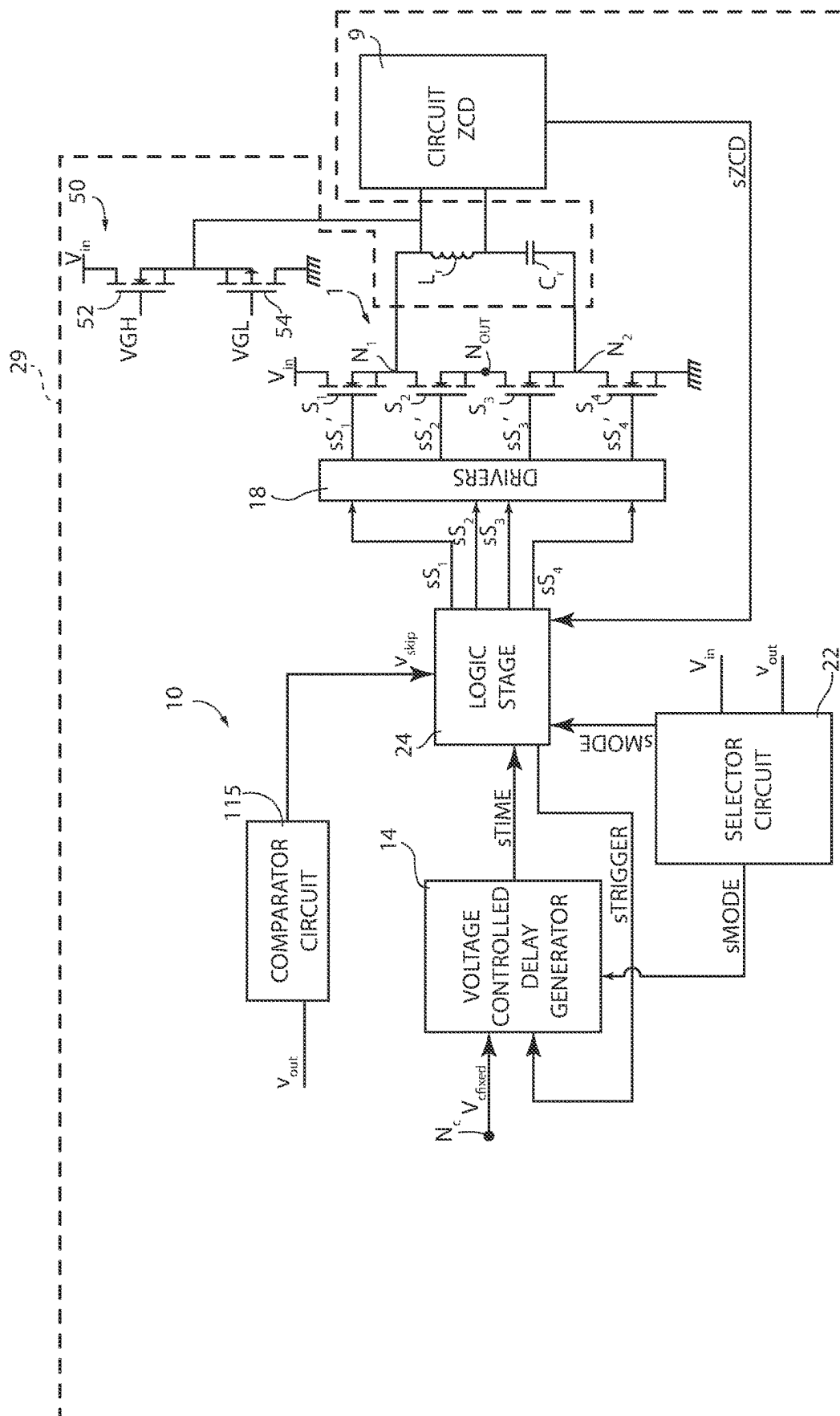


Fig. 9

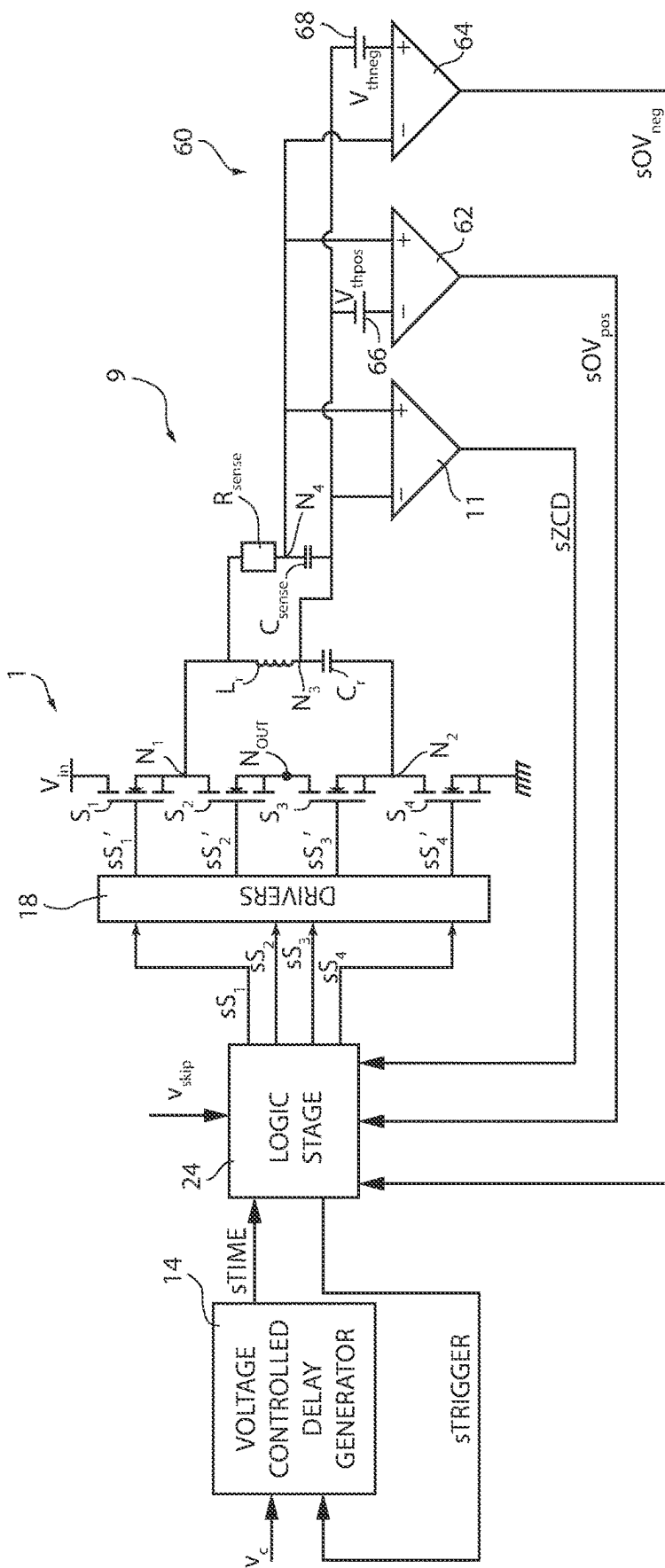


Fig. 10

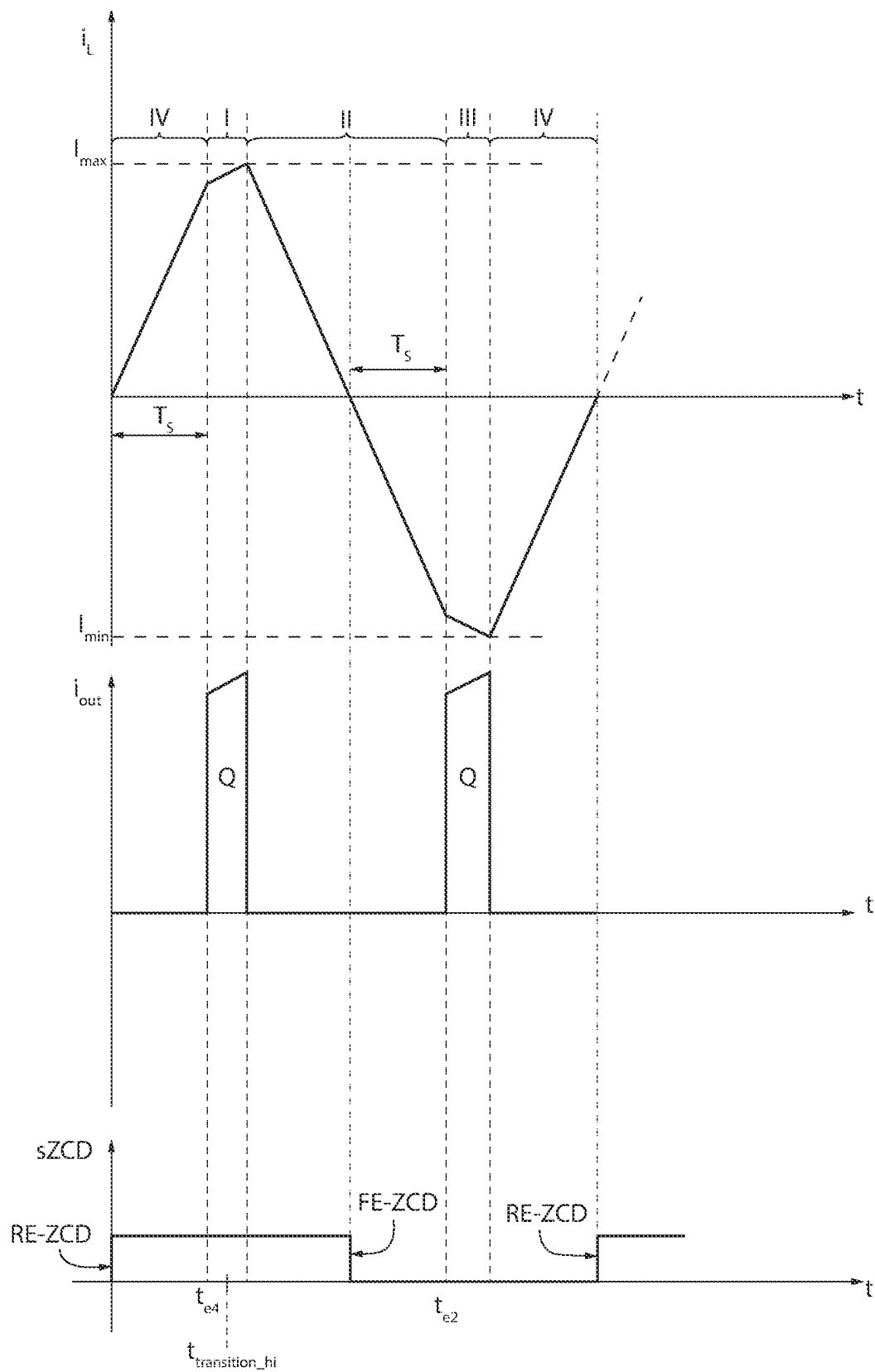


Fig. 11

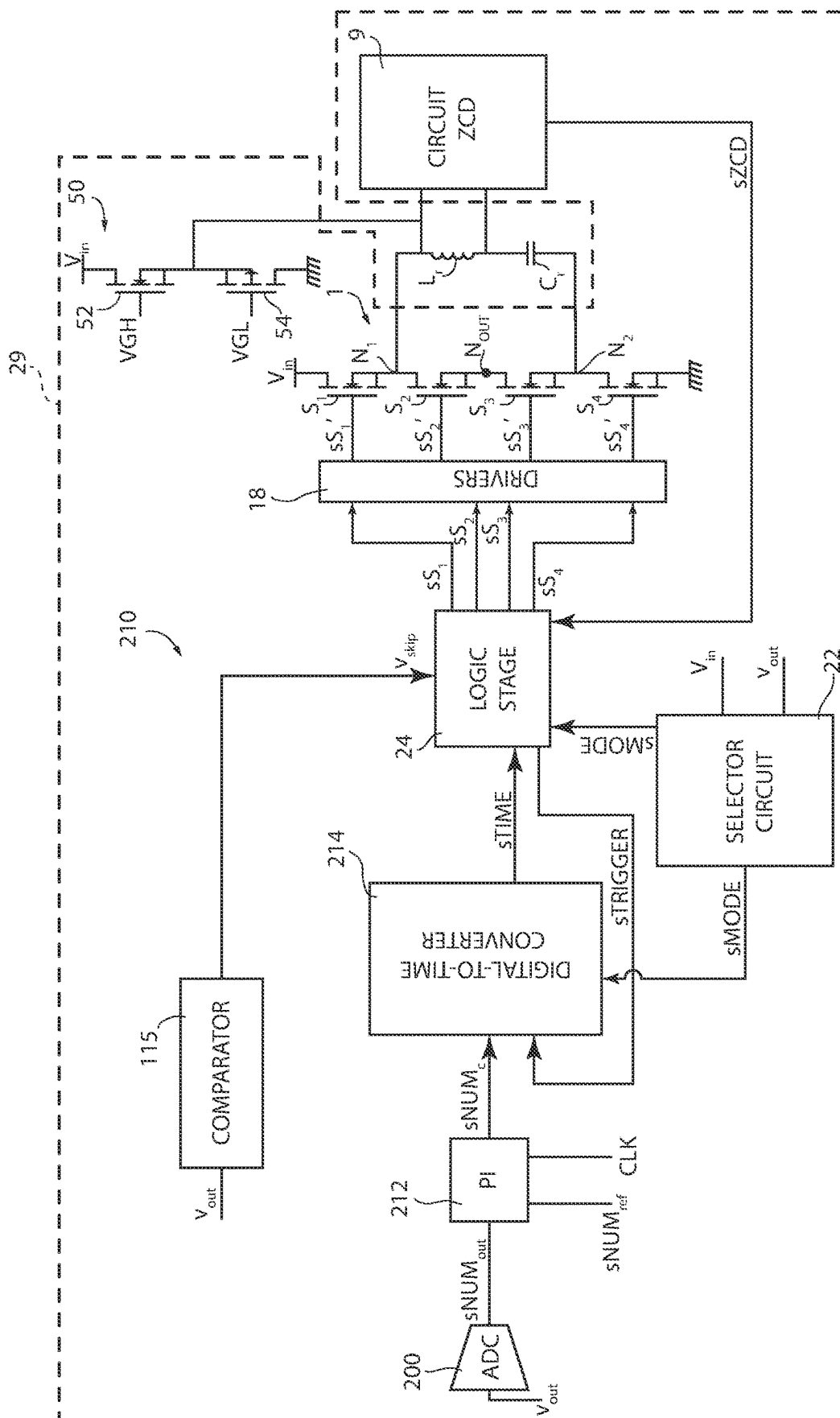


Fig. 12

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RESONANT SWITCHED CAPACITOR CONVERTER CONTROL MODULE AND METHOD WITH ZERO CURRENT SWITCHING CONTROL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of Italian Patent Application No. 102022000008339, filed on Apr. 27, 2022, which application is hereby incorporated herein by reference.

TECHNICAL FIELD

The disclosure relates to a control module for a resonant switched-capacitor converter with improved efficiency at low load and to a method for controlling the resonant switched-capacitor converter.

BACKGROUND

As it is known, the so-called switched-capacitor converters (SCCs) allow to regulate a direct-current (DC) output voltage V_{out} , starting from a DC input voltage V_{in} , by transferring energy (charge) between capacitors. Furthermore, among the switched-capacitor converters, there are known the so-called resonant switched-capacitor converters (reSCC), which provide for the presence of an inductor, which forms, along with a capacitor, a resonant circuit, thereby allowing to overcome some of the drawbacks of the purely capacitive switched-capacitor converters, such as the need for the input voltage V_{in} to be greater than a threshold (as an example, in the so-called 2:1 topology, the requirement $V_{in} > 2 \cdot V_{out}$ must be met). In addition, the presence of a resonant circuit allows to transfer the charge in an efficient way, because it allows to implement a zero-voltage switching of the transistors, i.e. it allows to reduce the V_{DS} voltage of the transistors during the switching. In addition, since the resonant circuit is driven at a frequency close to the resonance frequency, the inductor can have a very low inductance.

As an example, FIG. 1A shows a converter **1**, which is a resonant switched-capacitor DC-DC converter.

The converter **1** includes a first, a second, a third and a fourth transistor S_1 , S_2 , S_3 , S_4 , which are NMOS transistors.

The first, the second, the third and the fourth transistor S_1 - S_4 are connected in cascade. In particular, the drain terminal of the first transistor S_1 forms an input node N_{IN} , which is configured to be set to the input voltage V_{in} (referred to ground) whereas the source terminal of the first transistor S_1 forms a first intermediate node N_1 and is connected to the drain terminal of the second transistor S_2 . The input voltage V_{in} is generated by an input source **2**.

The source terminal of the second transistor S_2 forms an output node N_{OUT} and is connected to the drain terminal of the third transistor S_3 .

The source terminal of the third transistor S_3 forms a second intermediate node N_2 and is connected to the drain terminal of the fourth transistor S_4 , whose source terminal is connected to ground.

The converter **1** further includes a resonant circuit comprising an inductor L_r and a capacitor C_r , which are connected in series between the first and the second intermediate node N_1 , N_2 . The capacitance of the capacitor C_r may be comprised in the range 10 pF-10 μ F; the inductance of the inductor L_r may be comprised in the range 1 nH-10 μ H.

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The converter **1** further includes an output capacitor C_o , which is connected between the output node N_{OUT} and ground. In use, a load **3** is connected between the output node N_{OUT} and ground, in parallel to the output capacitor C_o . The average voltage on the output node N_{OUT} represents the average output voltage V_{out} .

In order to control the converter **1**, a control module (not shown) generates corresponding digital signals sS_1 , sS_2 , sS_3 , sS_4 . Though not shown, the gate terminals of the first, the second, the third and the fourth transistor S_1 - S_4 are controlled by corresponding signals which respectively depend on the signals sS_1 - sS_4 . In particular, by referring to an example to the first transistor S_1 , when the signal sS_1 is respectively equal to '1' or '0', the corresponding signal present on the gate terminal of the first transistor S_1 is respectively equal to a voltage V_{on1} or a voltage V_{off1} , which are such that the first transistor S_1 is respectively on or off; the same applies to the signals sS_2 , sS_3 , sS_4 and, respectively, the second, the third and the fourth transistor S_2 , S_3 , S_4 . Therefore, by adopting an integer index 'i' (with $i=1, 2, 3, 4$), when the signal sS_i is respectively equal to '1' or '0', the gate terminal of the i-th transistor S_i is set to a voltage which is respectively equal to V_{on_i} or V_{off_i} , the values V_{on_i} or V_{off_i} being such that the i-th transistor S_i is respectively on (i.e., the voltage gate-source is greater than the threshold) or off (i.e., the voltage gate-source is lower than the threshold). In addition, considered any of the signals sS_1 - sS_4 , the durations of the time periods in which the signal is equal to '1' or '0' are respectively designated by T_{ON} and T_{OFF} .

This having been said, the signal sS_2 is equal to the logic negation of the signal sS_1 ; the signal sS_4 is equal to the logic negation of the signal sS_3 . Therefore, the signals sS_1 and sS_2 form a first pair of signals sharing the same timing, whereas signals sS_3 and sS_4 form a second pair of signals sharing a same timing; the second pair of signals is time shifted with respect to the first pair of signals.

Based on the values assumed by the signals sS_1 - sS_4 , four phases are possible.

In particular, when the signals sS_1 and sS_3 are equal to '1', and thus signals sS_2 and sS_4 are equal to '0', the converter **1** operates in a first phase, shown in FIG. 1A (in which the transistors that are on and the path followed by the current are depicted in bold line). In the first phase, the first and the third transistor S_1 , S_3 are on, whereas the second and the fourth transistor S_2 , S_4 are off. Therefore, by referring to the current i_L to designate the current that flows through the inductor L_r , during the first phase the current i_L flows through the first and the third transistor S_1 , S_3 and is delivered to the output capacitor C_o and the load **3**.

When the signals sS_1 and sS_3 are respectively equal to '0' and '1', and thus the signals sS_2 and sS_4 are respectively equal to '1' and '0', the converter **1** operates in a second phase, shown in FIG. 1B. In the second phase, the first and the fourth transistor S_1 , S_4 are off, whereas the second and the third transistor S_2 , S_3 are on, thereby forming a first closed loop including the second and the third transistor S_2 , S_3 , the inductor L_r and the capacitor C_r . Therefore, during the second phase the current i_L recirculates in first closed loop, without being delivered to the output capacitor C_o and the load **3**.

When the signals sS_1 and sS_3 are equal to '0', and thus the signals sS_2 and sS_4 are equal to '1', the converter **1** operates in a third phase, shown in FIG. 1C. In the third phase, the first and the third transistor S_1 , S_3 are off, whereas the second and the fourth transistor S_2 , S_4 are on. Therefore, during the

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third phase, the current i_L flows through the second and the fourth transistor S_2 , S_4 and is delivered to the output capacitor C_o and the load **3**.

When the signals sS_1 and sS_3 are respectively equal to '1' and '0', and thus the signals sS_2 and sS_4 are respectively equal to '0' and '1', the converter **1** operates in a fourth phase, shown in FIG. 1D. In the fourth phase, the first and the fourth transistor S_1 , S_4 are on, whereas the second and the third transistor S_2 , S_3 are off, thereby forming a second closed loop including the input source **2**, the first and the fourth transistor S_1 , S_4 , the inductor L_r and the capacitor C_r . Therefore, during the fourth phase the current i_L recirculates in second closed loop, without being delivered to the output capacitor C_o and the load **3**.

In practice, the second and the fourth phase allow to adjust the amount of current delivered during the first and the third phase and to regulate any average output voltage V_{out} lower than the input voltage V_{in} .

A known control strategy provides for controlling the converter **1** so that it operates according to a sequence of phases formed by the iteration of a succession of the first, the second, the third and the fourth phase, as explained in greater detail in the following. Furthermore, the durations of the third and the fourth phase are respectively equal to the durations of the first and the second phase; in addition, by defining as switching period T_{SW} the aggregated duration of the first, the second, the third and the fourth phase, it is known to provide for two symmetrical semi-periods of duration $T_{SW}/2$, each semi-period consisting of two corresponding phases. In particular, a first semi-period consists of the first and the second phase and has duration equal to $T_{SW}/2$; a second semi-period consists of the third and the fourth phase and has duration equal to $T_{SW}/2$. Furthermore, hereinafter reference is made to the frequencies f_{SW} and f_{RES} to respectively designate the quantity $1/T_{SW}$ (also known as switching frequency) and the resonance frequency of the resonant circuit formed by the inductor L_r and the capacitor C_r .

The control module implements a control loop that adjusts the duration of the first phase or the second phase (and thus also of the third phase or the fourth phase) as a function of the difference between an output voltage v_{out} , which is the instantaneous voltage on the output node N_{OUT} , and a reference voltage V_{ref} , this difference being also known as error. To this regard, the control loop has a bandwidth such that the abovementioned duration of the first or second phase depends on the difference between the average output voltage V_{out} (i.e., the average over a switching period T_{SW} of the output voltage v_{out}) and the reference voltage V_{ref} .

A so-called zero-crossing detector (ZCD) circuit is implemented so as to determine the duration of the second phase (and thus also of the fourth phase).

In detail, as shown in FIG. 2 (wherein, for the sake of simplicity, the symbols showing the first, the second, the third and the fourth transistor S_1 , S_2 , S_3 , S_4 don't include the corresponding body diodes), the ZCD circuit (designated by **9**) may include a sense resistor R_{sense} , a sense capacitor C_{sense} and an amplifier **11**.

The sense resistor R_{sense} and the sense capacitor C_{sense} are connected in series so as to form a sense circuit **13**, which is connected in parallel to the inductor L_r . Furthermore, the terminals of the sense capacitor C_{sense} are respectively connected to the positive and negative input terminals of the amplifier **11**, which is configured to generate on its output terminal a signal $sZCD$.

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In use, under the assumption that the time constant of the sense capacitor C_{sense} is much greater than the time constant T_{SW} (with $T_{SW}=1/(2\pi*f_{SW})$), the sense circuit **13** is subjected, to a first approximation, to the same voltage of the inductor L_r (hereinafter referred to as v_L) therefore, the sense capacitor C_{sense} is charged by a current which is approximately equal to v_L/R_{sense} , which is integrated by the sense capacitor C_{sense} ; therefore, the voltage on the sense capacitor C_{sense} has the same trend over time as the current i_L . In practice, the voltage on the sense capacitor C_{sense} emulates the profile of the current i_L ; therefore, the signal $sZCD$ is equal to '1' or '0' when the current i_L is respectively positive or negative. Furthermore, whenever the current i_L crosses the zero, the signal $sZCD$ features a corresponding edge.

As an alternative (not shown), the ZCD circuit may be implemented in a manner such that it senses the current flowing through the transistors of the converter **1**.

Irrespective of the practical implementation of the ZCD circuit, the known control strategies provide for controlling the converter **1** in alternative ways, depending on whether $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$.

In particular, in case $V_{out} < V_{in}/2$, the control strategy provides for controlling the converter **1** so that it operates according to a sequence of phases formed by the iteration of the succession of the first, the second, the third and the fourth phase, such a sequence of phases starting with the first phase, as shown in FIG. 3A, which shows the plots over time of the current i_L , of the signal ZCD, of the signal sS_1 (coinciding with the signal $\overline{sS_2}$, i.e. with the logic negation of the signal sS_2) and of the signal sS_3 (coinciding with the signal sS_4 , i.e. with the logic negation of the signal sS_4).

In detail, by referring to the durations $T_{S'}$ and T_S to designate respectively the durations of the first and the second phase (and thus also, respectively, of the third and the fourth phase), the control module imposes the duration $T_{S'}$ as a function of the difference between the output voltage v_{out} (more precisely, the average output voltage V_{out}) and the reference voltage V_{ref} as mentioned before.

Furthermore, by assuming that $f_{SW} \gg f_{RES}$ (e.g., $f_{SW} \gg 10*f_{RES}$) and that, in view of the symmetry of the first, the second, the third and the fourth phase, at steady state the voltage on the capacitor C_r is approximately equal to $V_{in}/2$, it follows that, to a first approximation, during any first phase the current i_L rises, starting from zero, with a slope equal to $[(V_{in}/2) - V_{out}]/L$, with L representing the inductance of the inductor L_r . At the end of the first phase, the current i_L assumes a maximum positive value. Furthermore, during any first phase, a charge Q is transferred from the output node N_{OUT} to the load **3** and the output capacitor C_o ; in particular, by referring to i_{out} to designate the instantaneous current coming from the output node N_{OUT} and directed towards the load **3** and the output capacitor C_o , the current i_{out} is equal the current i_L .

During any second phase, the current i_L decreases with a slope equal to $-V_{in}/(2L)$, until it reaches the zero value. The second phase precisely ends when the current i_L reaches the zero value, such an event being detected by the ZCD circuit **9**.

During any third phase, the current i_L further decreases (i.e., it becomes negative) with a slope equal to $[V_{out} - (V_{in}/2)]/L$. At the end of the third phase, the current i_L assumes a maximum negative value. Furthermore, during any third phase, the charge Q is transferred to the load **3** and the output capacitor C_o ; in particular, the current i_{out} is equal to $-i_L$.

During any fourth phase, the current i_L rises, starting from the maximum negative value, with a slope equal to $V_{in}/(2L)$, until it reaches the zero value. The fourth phase precisely

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ends when the current i_L reaches the zero value, such an event being detected by the ZCD circuit 9. The end of the fourth phase coincides with the start of the following first phase.

In greater detail, by assuming $\alpha=2\pi*T_S/T_{SW}$, any second or fourth phase corresponds to a phase angle α , whereas any first or third phase corresponds to a phase angle equal to $\pi-\alpha$. Furthermore, the charge transferred to the load 3 and the output capacitor C_o during the first and the third phase, divided by T_{SW} , is equal to the current I_{out} , i.e. to the average of the current i_{out} over a switching period T_{SW} .

In particular, in case $V_{out}>V_{in}/2$, the control strategy provides for controlling the converter 1 so that it operates according to a sequence of phases formed by the iteration of the succession of the fourth, the first, the second and the third phase, as shown in FIG. 3B, which shows the plots over time of the same quantities shown in FIG. 3A.

In detail, the control module imposes the duration T_S of the second and fourth phases, as a function of the difference between the output voltage v_{out} (more precisely, the average output voltage V_{out}) and the reference voltage V_{ref} .

Furthermore, to a first approximation, during any fourth phase the current i_L rises, starting from zero, with a slope equal to $V_{in}/(2L)$. At the end of the fourth phase, the current i_L assumes a maximum positive value.

During any first phase, the current i_L decreases with a slope equal to $[(V_{in}/2)-V_{out}]/L$, until it reaches the zero value. The first phase precisely ends when the current i_L reaches the zero value, such an event being detected by the ZCD circuit 9. Furthermore, during any first phase, the charge Q is transferred to the load 3 and the output capacitor C_o ; in particular, the current i_{out} is equal to the current i_L .

During any second phase, the current i_L further decreases (i.e., it becomes negative) with a slope equal to $-V_{in}/(2L)$. At the end of the second phase, the current i_L assumes a maximum negative value.

During any third phase, the current i_L rises, starting from the maximum negative value, with a slope equal to $[V_{out}-(V_{in}/2)]/L$, until it reaches the zero value. The third phase precisely ends when the current i_L reaches the zero value, such an event being detected by the ZCD circuit 9. Furthermore, during any third phase, the charge Q is transferred to the load 3 and the output capacitor C_o ; in particular, the current i_{out} is equal to $-i_L$. The end of the third phase coincides with the start of the following fourth phase.

In greater detail, by assuming $\beta=2\pi*T_S/T_{SW}$, any second or fourth phase corresponds to a phase angle β , whereas any first or third phase corresponds to a phase angle equal to $\pi-\beta$. Furthermore, the charge transferred to the load 3 and the output capacitor C_o during the first phase and the third phase, divided by T_{SW} , is equal to the current I_{LOAD} , i.e. to the average current absorbed by the load 3 during the switching period T_{SW} .

In practice, if $V_{out}<V_{in}/2$, the control module sets the duration T_S' , whereas the duration T_S depends on the signal sZCD; on the contrary, if $V_{out}>V_{in}/2$, the control module sets the duration T_S , whereas the duration T_S' depends on the signal sZCD. In both cases, the durations T_S and T_S' are adjusted so as to keep the average output voltage V_{out} at a desired value.

This having been said, it is possible to demonstrate that, to a first approximation, the following equation applies:

$$I_{out}=V_{in}*\tan[(\omega_r*T_{SW})/4]/(Z_r*T_{SW})*T_S \quad (1)$$

wherein $Z_r=(L_r/C_r)^{1/2}$ and $\omega_r=1/(L_r*C_r)^{1/2}$

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Furthermore, based on geometrical considerations, it is possible to demonstrate that:

$$\alpha=\pi*[(V_{in}/2)-V_{out}]/(V_{in}-V_{out}) \quad (2)$$

$$\beta=\pi*[V_{out}-(V_{in}/2)]/V_{out} \quad (3)$$

It is thus apparent that the angles α and β only depend on the input voltage V_{in} and the average output voltage V_{out} . Therefore, also the ratio T_S/T_{SW} is constant. Furthermore, assuming $f_{SW}>>f_{RES}$, it is possible to demonstrate that, to a first approximation:

$$I_{out}=(V_{in}*\pi*f_{RES})/(2*Z_r)*T_S=V_{in}/(4*L)*T_S \quad (4)$$

Since at steady state the current I_{out} is equal to the current I_{LOAD} , and considering that $T_S/T_{SW}=T_S*f_{SW}=\text{constant}$, it follows that the switching frequency f_{SW} is directly proportional to the inverse of the current I_{LOAD} . Put in other words, when the current I_{LOAD} decreases, the switching frequency f_{SW} has to increase in order to regulate the average output voltage V_{out} . That leads to a loss of efficiency, when the current I_{LOAD} is low, because of the losses caused by the switching.

In order to overcome the above loss of efficiency, it is known to control the converter 1 so as force the converter 1 to work at a constant frequency f_{SW} , i.e. so as to ignore the signal sZCD. However, in this case, when the current I_{LOAD} approaches to zero, the second and the fourth phase are skipped, therefore during any first or third phase, charge is transferred to the load 3 and the output capacitor C_o and then taken back; as a consequence, the efficiency is low.

As an alternative, it is possible to control the converter 1 so as force the converter 1 to work at a constant frequency f_{SW} much lower than f_{RES} . Also, in this case, the second and the fourth phase are skipped, and a fixed amount of charge is delivered to the load 3 for each pulse, the repetition time of the pulses being adjusted as a function of the current I_{LOAD} . However, in this case the converter 1 operates in the capacitive region (i.e., the inductor L_r is substantially equivalent to a short circuit), so it is possible to provide power (i.e., to transfer a positive charge) to the load 3 only if $V_{out}<V_{in}/2$.

Various embodiments thus provide a module for controlling a resonant switched-capacitor converter that will enable the drawbacks of the prior art to be overcome at least in part.

SUMMARY

According to various embodiments, a control module and a method for controlling a resonant switched-capacitor converter are provided, as defined in the annexed claims.

BRIEF DESCRIPTION OF THE DRAWINGS

For a better understanding of the present invention, preferred embodiments thereof will now be described purely by way of non-limiting example, with reference to the attached drawings, wherein:

FIGS. 1A-1D show circuit schemes of a portion of a converter, in different operating phases;

FIG. 2 shows a circuit scheme of a portion of a converter; FIGS. 3A and 3B show time plots of a current and of signals;

FIGS. 4, 9, 10, 12 show circuit schemes of control modules;

FIGS. 5, 7 and 11 show time plots of signals generated in the control module;

FIG. 6 shows a time plot of a voltage generated by the converter; and

FIGS. 8A and 8B show time plots of currents, during corresponding operating conditions.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

FIG. 4 shows a control module 10, which controls the converter 1 shown in FIGS. 1A-1D. For the sake of simplicity, FIG. 4 does not show the load 3 and the output capacitor C_c .

In detail, the control module 10 comprises the ZCD circuit 9, an operational transconductance amplifier 12, a voltage controlled delay generator 14, a driver stage 18 and a compensation network 15 including a resistor R_c and a capacitor C_c . In addition, the control module 10 comprises a clamping circuit 20, a selector circuit 22 and a logic stage 24. Though not shown, the control module 10 may further include a reference generator configured to generate an output reference voltage V_{out_ref} . The operational transconductance amplifier 12 is of the single-ended type.

Though not shown, the negative input terminal of the operational transconductance amplifier 12 is connected to the output node N_{OUT} , in order to receive the output voltage v_{out} . The positive input terminal of the operational transconductance amplifier 12 is set to the output reference voltage V_{out_ref} . Therefore, an error voltage $v_{err} = V_{out_ref} - v_{out}$ is present on the input of the operational transconductance amplifier 12.

The output terminal of the operational transconductance amplifier 12 forms a control node N_c . The compensation network 15 is connected between the control node N_c and ground. In particular, a first terminal of the resistor R_c is connected to the control node N_c ; a second terminal of the resistor R_c is connected to a first terminal of the capacitor C_c , whose second terminal is connected to ground. Hereinafter, reference is made to the control voltage v_c to designate the voltage on the control node N_c .

As an example, the capacitance of the capacitor C_c may be comprised in the range 1 pF-1 nF; the resistance of the resistor R_c may be comprised in the range 1 k Ω -1 M Ω .

Assuming a very high value of the output resistance of the operational transconductance amplifier 12, this latter injects into (or drains from) the control node N_c a current i_{OTA} which is directly proportional to the error voltage v_{err} ; as an example, the current i_{OTA} is respectively injected or drained, if the error voltage v_{err} is respectively positive or negative. In particular, $i_{OTA} = gm_{OTA} * v_{err}$, wherein gm_{OTA} is the gain of the operational transconductance amplifier 12.

From a practical point of view, and to a first approximation, the operational transconductance amplifier 12 and the compensation network 15 form a proportional integrative controller, because the dependence of the control voltage v_c on the current i_{OTA} , and thus on the error voltage v_{err} , is approximately of the proportional integrative type. From a physical standpoint, this dependence is explained here below with reference, as an example, to the case $i_{OTA} > 0$.

In detail, at low frequencies, i.e. in case of slowly varying components of the error voltage v_{err} , the capacitor C_c dominates the behaviour of the compensation network 15; therefore, the current i_{OTA} is integrated by the capacitor C_c , thereby causing the control voltage v_c to rise. At high frequencies, i.e. in case of quickly varying components of the error voltage v_{err} , the capacitor C_c acts as a short circuit,

so the current i_{OTA} causes a voltage drop on the compensation network 15, which depends on the resistance of the resistor R_c .

The selector circuit 22 receives the input voltage V_{in} and the output voltage v_{out} . Furthermore, the selector circuit 22 compares the average output voltage V_{out} with $V_{in}/2$ and generates a signal sMODE, which is equal to '1' or '0', respectively when $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$.

The voltage controlled delay generator 14 has inputs respectively connected to the control node N_c , to receive the control voltage v_c , and to the logic stage 24, to receive a signal sTRIGGER generated by this latter. As shown in FIG. 5, the signal sTRIGGER may be a periodic pulsed signal, with duty cycle lower than 50%; in particular, the logic stage 24 generates a rising edge of the signal sTRIGGER whenever a rising or falling edge of the signal sZCD occur.

The voltage controlled delay generator 14 generates a signal sTIME. As an example, the voltage controlled delay generator 14 may be configured so as to generate a falling edge of the signal sTIME at each rising edge of the signal sTRIGGER, as shown in FIG. 5.

Furthermore, the voltage controlled delay generator 14 may be configured so as to charge, after each rising edge of the signal sTRIGGER, a respective capacitor (not shown) with a fixed current, and to generate a rising edge of the signal sTIME when the voltage on the capacitor (designated by v_{RAMP} in FIG. 5) reaches a threshold value linearly depending on (e.g., equal to) the control voltage v_c , thereby causing each rising edge of the signal sTIME to be delayed by a delay T^* from the preceding falling edge. To a first approximation, the delay T^* linearly depends on the control voltage v_c .

In practice, the signal sTIME represents the delay T^* , which assumes a corresponding value for each switching period T_{SW} . Furthermore, the fixed current that charges the abovementioned capacitor assumes a first or a second predetermined value, respectively when the signal sMODE is equal to '1' or '0', the first value being as an example lower than the second value. In this way, given any value of the control voltage v_c , the corresponding value of the delay T^* depends on whether $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$; in particular, given any value of the control voltage v_c , the value of the corresponding delay T^* is higher in case $V_{out} < V_{in}/2$. In this way, it is possible to use the same voltage controlled delay generator 14 to generate the correct values of the delay T^* in both the scenarios $V_{out} < V_{in}/2$ and $V_{out} > V_{in}/2$, instead of using two different voltage controlled delay generators. Furthermore, though not described, embodiments are possible in which the behaviour of voltage controlled delay generator 14 does not depend on the signal sMODE, i.e. on the scenario, in which case the circuit complexity may increase.

As an alternative, the voltage controlled generator 14 may be formed by a chain of inverters, whose supply voltage is equal to the control voltage v_c , so that the delay introduced by the inverter chain depends on the control voltage v_c . In this case, the delay decreases as the control voltage v_c increases, so the closed loop that controls the output voltage v_{out} may be modified so as to maintain a global negative feedback. In any case, the technical implementation of the voltage controlled delay generator 14 is irrelevant for the implementation of various embodiments.

In greater detail, when the control module 10 operates in a normal mode, the delay T^* represented by the signal sTIME is used by the logic stage 24 to set either the duration T_S' (if $V_{out} < V_{in}/2$) or the duration T_S (if $V_{out} > V_{in}/2$).

As an example, assuming that FIG. 5 refers to the case $V_{out} < V_{in}/2$ and assuming as a reference a time instant of any fourth phase, the logic stage 24 ends this fourth phase upon receiving a rising edge (designated by RE-ZCD in FIG. 5) of the signal sZCD and then implements a first phase (i.e., the digital signals sS₁, sS₂, sS₃, sS₄ assume corresponding values such that a first phase is carried out) for a time interval equal to the delay T* represented by the signal sTIME; put in other words, the first phase lasts until the reception of the first rising edge (designated by RE-TIME1 in FIG. 5) of the signal sTIME following the rising edge RE-ZCD of the signal sZCD. Then, the logic stage 24 implements a second phase, until it receives a falling edge of the signal sZCD, designated by FE-ZCD in FIG. 5. Then, the logic circuit 24 implements a third phase for a time interval equal to the delay T* represented by the signal sTIME (to a first approximation, this value hasn't changed with respect to the preceding first phase), i.e. it implements a third phase lasting till the reception of the first rising edge (designated by RE-TIME2 in FIG. 5) of the signal sTIME following the falling edge FE-ZCD of the signal sZCD. Finally, the logic stage 24 implements a new fourth phase, until the reception of the following rising edge of the signal sZCD (not shown in FIG. 5).

The same considerations apply in case $V_{out} > V_{in}/2$. In particular, assuming as a reference a time instant of any third phase, the logic stage 24 ends this third phase upon receiving the rising edge RE-ZCD of the signal sZCD and then implements a fourth phase for a time interval equal to the delay T* represented by the signal sTIME; put in other words, the fourth phase lasts till the reception of the rising edge RE-TIME1 of the signal sTIME. Then, the logic stage 24 implements a first phase, until it receives the falling edge FE-ZCD of the signal sZCD. Then, the logic circuit 24 implements a second phase for a time interval equal to the delay T* represented by the signal sTIME, i.e. it implements a second phase lasting till the reception of the rising edge RE-TIME2 of the signal sTIME. Finally, the logic stage 24 implements a new third phase, till the reception of the following rising edge of the signal sZCD (not shown in FIG. 5).

The clamping circuit 20 comprises a first, a second and a third clamping transistor 100, 102, 104, an operational amplifier 106, a clamping resistor 108 and a comparator circuit 110.

The first clamping transistor 100 is a NMOS transistor, with source terminal connected to the control node N_c. The second and the third clamping transistor 102, 104 are PMOS transistors and form a current mirror. In particular, the gate and drain terminals of the second clamping transistor 102 are connected to the drain terminal of the first clamping transistor 100; in addition, the gate terminal of the second clamping transistor 102 is connected to the gate terminal of the third clamping transistor 104. The source terminal of the second clamping transistor 102 is set to a supply voltage V_{dd}, which may equal, as an example, to the input voltage V_{in}.

The source terminal of the third clamping transistor 104 is set to the supply voltage V_{dd}. The drain terminal of the third clamping transistor 104 is connected to a first terminal of the clamping resistor 108, whose second terminal is connected to ground. The first terminal of the clamping resistor 108 forms a clamping node N_{clamp}; hereinafter reference will be made to the voltage v_{clamp} to designate the voltage on the clamping node N_{clamp}.

The negative input terminal of the operational amplifier 106 is connected to the control node N_c. The positive input

terminal of the operational amplifier 106 is set to a DC voltage V_{cmin}. The output terminal of the operational amplifier 106 is connected to the gate terminal of the first clamping transistor 100.

The input of the comparator circuit 110 is connected to the clamping node N_{clamp}, whereas the output is connected to the logic stage 24.

In detail, the comparator circuit 110 is a hysteretic comparator, which is configured to compare the voltage v_{clamp} with a first and a second threshold V_{thres_max}, V_{thres_min}, with $V_{thres_max} > V_{thres_min}$. Furthermore, the comparator circuit 110 generates on its output a signal v_{skip}.

As an example, the comparator circuit 110 may generate a rising edge (i.e., a transition from '0' to '1') of the signal v_{skip} whenever the voltage v_{clamp} overcomes the first threshold V_{thres_max}; in addition, the comparator circuit 110 may generate a falling edge (i.e., a transition from '1' to '0') of the signal v_{skip} whenever the voltage v_{clamp} drops below the second threshold V_{thres_min}.

In the following, reference will be made to the voltage v_{dc} to designate the voltage applied at steady-state by the operational transconductance amplifier 12 on the control node N_c when v_{out} = V_{out_ref}. Furthermore, it will be assumed that, in order to have, at steady-state, I_{out} = I_{LOAD}, the control voltage v_c has to be greater than the voltage v_{dc}; therefore, at steady-state the average output voltage V_{out} is lower than the output reference voltage V_{out_ref} (though, to a first approximation it may be assumed, for the sake of simplicity, $V_{out} = V_{out_ref}$ assuming a high loop gain and neglecting the ripple on the output voltage v_{out}). However, it has to be noted that, from the standpoint of the functioning of the control module, nothing would change if, at steady-state, the average output voltage V_{out} was greater than the output reference voltage V_{out_ref}; also, in this case, it would be possible to assume $V_{out} = V_{out_ref}$.

Furthermore, by referring to the current I_{LOAD_MIN} to indicate the value of the current I_{LOAD} such that the control voltage v_c is equal to the voltage V_{cmin}, if I_{LOAD} > I_{LOAD_MIN}, the control voltage v_c is higher than the voltage V_{cmin} and such that the corresponding value of the delay T* causes the current I_{out} to be equal to the current I_{LOAD}. According to whether the voltage V_{cmin} is lower or greater than the voltage v_{dc}, at steady-state the average voltage V_{out} is respectively greater or lower than the output reference voltage V_{out_ref}; however, as explained above, in both cases it may be assumed $V_{out} = V_{out_ref}$ at steady-state.

Since the control voltage v_c is higher than the voltage V_{cmin}, the first clamping transistor 100 is off; the clamping circuit 20 is thus decoupled from the control node N_c. Furthermore, no current flows in the current mirror formed by the second and the third clamping transistor 102, 104, thus the voltage v_{clamp} is equal to zero. In addition, the signal v_{skip} is equal to '0', and the logic stage 24 generates the signals sS₁, sS₂, sS₃ and sS₄ based on the signal sTIME, the signal sMODE and the signal sZCD, as shown in FIG. 3A or 3B and 5. As an example, if sMODE = '1' (i.e., $V_{out} < V_{in}/2$), the logic stage 24 generates the signals sS₁, sS₂, sS₃ and sS₄ as shown in FIG. 3A, so that the duration T_{S'} is equal to the delay T* represented by the signal sTIME. If sMODE = '0' (i.e., $V_{out} > V_{in}/2$), the logic stage 24 generates the signals sS₁, sS₂, sS₃ and sS₄ as shown in FIG. 3B, so that the duration T_S is equal to the delay T* represented by the signal sTIME.

The driver stage 18 is connected to the logic stage 24 and to the first, the second, the third and the fourth transistor S₁, S₂, S₃, S₄.

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In detail, the driver stage **18** receives the signals sS_1 , sS_2 , sS_3 and sS_4 and generates a corresponding first gate signal sS_1' , a corresponding second gate signal sS_2' , a corresponding third gate signal sS_3' and a corresponding fourth gate signal sS_4' , which are respectively applied to the gate terminals of the first, the second, the third and the fourth transistor S_1 , S_2 , S_3 , S_4 .

In greater detail, by adopting an integer index 'i' (with $i=1, 2, 3, 4$), when the signal sS_i is respectively equal to '1' or '0', the corresponding gate signal sS_i' is respectively equal to a voltage V_{on_i} or a voltage V_{off_i} , the voltages V_{on_i} and V_{off_i} being such that the i-th transistor S_i is respectively on or off.

This having been said, in case the current I_{LOAD} starts to decrease with respect to the abovementioned steady-state, the average output voltage V_{out} tends to rise (for the sake of simplicity, to a first approximation it may be assumed that such a rise is negligible, given the high gain of the control loop), so as to reduce the control voltage v_c and thus reduce the duration T_S' (if $V_{out} < V_{in}/2$) or the duration T_S (if $V_{out} > V_{in}/2$), in order to adapt, in both cases, the current I_{out} to the current I_{LOAD} . In practice, as shown in FIG. 6 (left part), if the current I_{LOAD} is equal to a value I_{LOAD_MIN} , the average output voltage V_{out} is still approximately equal to the output reference voltage V_{out_ref} .

If the current I_{LOAD} drops below the value I_{LOAD_MIN} , the control voltage v_c tends to drop below the voltage V_{cmin} , thereby causing the operational amplifier **106** to switch, thereby turning on the first clamping transistor **100** and coupling the clamping circuit **20** to the control node N_c .

In practice, the voltage V_{cmin} defines the minimum values admissible for the durations T_S' (if $V_{out} < V_{in}/2$) or T_S (if $V_{out} > V_{in}/2$), hereinafter referred to as T_{Smin}' or T_{Smin} , which are respectively equal to the two values of the delay T^* represented by the signal $sTIME$ when the control voltage v_c is equal to the voltage V_{cmin} , in the cases $V_{out} < V_{in}/2$ and $V_{out} > V_{in}/2$.

Since the clamping circuit **20** is now coupled to the control node N_c , the drop of the control voltage v_c is temporarily stopped, i.e. the control voltage v_c is temporarily stuck to the voltage V_{cmin} . That implies that the current I_{out} becomes higher than the current I_{LOAD} . Therefore, the average output voltage V_{out} rises above the output reference voltage V_{out_ref} , as shown FIG. 6 (right part), thereby causing an inversion of the sign of the error voltage v_{err} . Therefore, the operational transconductance amplifier **12** attempts to decrease the control voltage v_c , by draining current from the control node N_c , this current, hereinafter referred to as the current i_{clamp} , being provided by the clamping circuit **20**, so as to allow the control voltage v_c to remain stuck at the voltage V_{cmin} . In practice, $i_{clamp} = -i_{OTA} = -gm_{OTA} * v_{err}$.

In greater detail, the current i_{clamp} flows through the first and the second clamping transistor **100**, **102** and it is mirrored by the current mirror, so that the same current i_{clamp} flows through the third clamping transistor **104**; therefore, the voltage v_{clamp} rises over time, until it reaches the first threshold V_{thres_max} , thereby causing the comparator circuit **110** to generate a rising edge of the signal v_{skip} , as shown in FIG. 7, which refers as an example to the case $V_{out} > V_{in}/2$. The time instant in which the rising edge of the signal v_{skip} occurs is designated by t_{hi} .

In practice, the voltage v_{clamp} (not shown in FIG. 7) reaches the first threshold V_{thres_max} when the current i_{clamp} reaches a corresponding threshold I_{clamp_hi} . Furthermore, the first threshold V_{thres_max} relating to the voltage v_{clamp} corresponds to a first output threshold V_{out_hi} relating to the

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output voltage v_{out} , i.e., when the voltage v_{clamp} reaches the first threshold V_{thres_max} , the output voltage v_{out} reaches the first output threshold V_{out_hi} . To this regard, FIGS. 6 and 7 refer to two different examples. In addition, in FIG. 6, the time instant in which the output voltage v_{out} reaches the first output threshold V_{out_hi} is designated by t_x ; furthermore, FIG. 6 shows the trend of the output voltage v_{out} when passing from $I_{LOAD} = I_{LOAD_MIN}$ to $I_{LOAD} < I_{LOAD_MIN}$, up to the time instant t_x . FIG. 7 refers to the case $I_{LOAD} < I_{LOAD_MIN}$; furthermore, in FIG. 7 the time instant in which the output voltage v_{out} reaches the first output threshold V_{out_hi} is designated by t_{hi} . Hereinafter, reference will be made to the example of FIG. 7 to explain the events occurring after the time instant in which the output voltage v_{out} reaches the first output threshold V_{out_hi} .

In detail, it is possible to demonstrate that:

$$V_{out_hi} = V_{out_ref} + I_{thres_max} / gm_{OTA} \quad (5)$$

After the signal v_{skip} goes to '1', the logic stage **24** enters into a pulse skipping mode, as explained here below, with reference to the example shown in FIG. 7, in which the time instant t_{hi} falls into a first phase, hereinafter referred to as the initial phase.

Since the signal v_{skip} has gone to '1', at the end of the initial phase, which, as explained above, is marked by the falling edge FE-ZCD of the signal $sZCD$ (which occurs at a time instant t_{ZCD}), the logic stage **24** stops the execution of the phase sequence, which will be resumed later on. Therefore, after the time instant t_{hi} , the output voltage v_{out} rises only until (approximately) the end of the abovementioned initial phase; during the time interval comprised between the time instant t_{hi} and the end of the abovementioned initial phase, the current i_{clamp} may assume values higher than the first threshold I_{clamp_hi} .

In order to stop the execution of the phase sequence, the logic stage **24** may, as an example, set all the signals sS_1 , sS_2 , sS_3 equal to '0', in order to temporarily turn off the first, the second and the third transistor S_1 , S_2 , S_3 . Put in other words, the output node N_{OUT} is set to a high impedance condition.

Since the execution of the phases is suspended, the output voltage v_{out} decreases, starting approximately from the end of the abovementioned initial phase and with a slope depending on the ratio I_{LOAD}/C_o , so also the current i_{clamp} and the voltage v_{clamp} decrease. At a time instant t_{lo} , the current i_{clamp} reaches a corresponding second threshold I_{clamp_lo} , thus the voltage v_{clamp} reaches the second threshold V_{thres_min} and the output voltage v_{out} reaches a second output threshold V_{out_lo} , which is lower than the first output threshold V_{out_hi} and greater than the output reference voltage V_{out_ref} .

In particular, it is possible to demonstrate that:

$$V_{out_lo} = V_{out_ref} + I_{thres_min} / gm_{OTA} \quad (6)$$

At the time instant t_{lo} , since the voltage v_{clamp} has reached the second threshold V_{thres_min} , the signal v_{skip} goes to '0', which in turn causes the logic stage **24** to resume the execution of the phase sequence; put in other words, the control module **10** temporarily resumes to operate in normal mode.

As an example, referring to FIG. 7, after the time instant t_{lo} , the logic stage **24** implements the second phase (i.e., it resumes to generate the signals sS_1 , sS_2 , sS_3 , sS_4 in a manner such that a second phase is implemented) and then a third phase. Furthermore, the time interval extending between the time instant t_{ZCD} and the time instant t_{lo} represents a pulse-skipping period T_{skip} .

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During the second phase that follows the time instant t_{lo} , the output voltage v_{out} and the current i_{clamp} continue to decrease, i.e. they assume values respectively lower than the second output threshold V_{out_lo} and the second threshold I_{clamp_lo} , because no charge is delivered to the load **3**, nor to the output capacitor C_o . In particular, at the end of the second phase, the output voltage v_{out} is equal to a minimum value V_{out_min} . In case $V_{out_min} > V_{out_ref}$, the current i_{clamp} remains different from zero, in which case the control voltage v_c stays stuck to the voltage v_{cmin} . On the contrary, if $V_{out_min} \leq V_{out_ref}$, the current i_{clamp} reaches the zero value and the control voltage v_c is no more stuck to the voltage v_{cmin} , so it may temporarily rise, until when the output voltage v_{out} reaches the output reference voltage V_{out_ref} , thereby causing a new clamping of the control voltage v_c to the voltage v_{cmin} .

During the following third phase, and under the assumption that the current I_{LOAD} has not changed (i.e., the current I_{LOAD} is still lower than I_{LOAD_MIN}), the output voltage v_{out} and the current i_{clamp} rise. In particular, at a time instant t_{hi}' , it occurs $i_{clamp} = I_{clamp_hi}$, $v_{clamp} = V_{thres_max}$, and the output voltage v_{out} reaches again the first output threshold V_{out_hi} ; therefore, at the time instant t_{hi}' , the signal v_{skip} goes to '1', the logic stage **24** enters again into the pulse skipping mode and the steps previously described are iterated: upon terminating the third phase (i.e., the new initial phase), as signalled by the rising edge RE-ZCD of the signal sZCD that follows the time instant t_{hi}' , which occurs at a time instant t_{ZCD}' , the execution of the phases is suspended, until when the voltage v_{clamp} drops again below V_{thres_min} (not shown in FIG. 7).

In practice, the fourth and the first phase shown in FIG. 7 form a first semi-pulse, whereas the second and the third phase form a second semi-pulse. The first and the second semi-pulse are spaced apart in time, the time separation being inversely proportional to the current I_{LOAD} . The lower the value of the current I_{LOAD} , the higher is the time separation between the first and the second semi-pulse; the same considerations apply to possible semi-pulses (not shown in FIG. 7) that follow the second semi-pulse, so that the converter **1** is operated at a high effective switching period T_{Sweff} , meant as the time period during which a pair of consecutive semi-pulses occurs, thereby reducing the loss of efficiency due to switching losses. In facts, $T_{Sweff} = T_{SW} + 2 \cdot T_{skip}$, so the effective switching frequency and the related switching losses are reduced. Furthermore, the following equation applies:

$$I_{out} = 2 \cdot Q / T_{Sweff} \quad (7)$$

Still with reference to FIG. 7, had the current I_{LOAD} risen to a value greater than I_{LOAD_MIN} before the execution of the abovementioned third phase, the current i_{clamp} and the output voltage v_{out} could not have reached, respectively, the threshold I_{clamp_hi} and the first output threshold V_{out_hi} ; therefore the signal v_{skip} wouldn't have gone to '1'; as a consequence, the execution of the phases would have continued without any further suspension, and the average output voltage V_{out} would have gradually dropped to (approximately) the output reference voltage V_{out_ref} , thereby causing the control voltage v_c to decouple from the voltage v_{cmin} .

In greater detail, though the embodiment of FIG. 4 is such that the time instants t_{hi} , t_{hi}' in which the signal v_{skip} goes to '1' fall within a first and a third phase, i.e. during phases in which charge is delivered to the output node N_{out} , variants are possible. In general, each of the time instants t_{hi} , t_{hi}' may occur during any phase, as an example because of delays

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introduced by the clamping circuit, which may be different from the proposed one. To this regard, as an example, instead of applying a comparator to the voltage v_{clamp} , i.e. to a proxy quantity of the current i_{clamp} , the output voltage v_{out} may be directly applied to a comparator with hysteresis and thresholds equals to the first and the second output thresholds V_{out_hi} , V_{out_lo} .

In general, assuming anytime instant $t_{transition_hi}$ in which the signal v_{skip} goes to '1', the logic stage **24** can suspend the execution of the phase sequence at the first edge (either rising or falling) of the signal sZCD that follows the time instant $t_{transition_hi}$. In facts, the execution of the phase sequence can be suspended when the current i_L is null. Furthermore, embodiments are possible in which the logic stage **24** suspends the execution of the phase sequence at the k-th (with k equal to a positive integer number greater than one) edge (either rising or falling) of the signal sZCD that follows the time instant $t_{transition_hi}$; however, the greater is the number k, the greater is the ripple on the output voltage v_{out} . To this regard, by suspending the execution at the first edge (either rising or falling) following the occurrence of $v_{out} > V_{out_hi}$, it is possible to reduce the ripple ΔV_{pp} on the output voltage v_{out} to (approximately) Q/C_o . Furthermore, in general the signal sZCD may be disregarded by the logic stage **24** during any period T_{skip} .

As an example, FIG. 8A refers to the case $V_{out} > V_{in}/2$, with time instant $t_{transition_hi}$ (only one indicated in FIG. 8A) falling within a first phase. In this embodiment, the execution of the phase sequence is suspended after the execution of a whole pulse, i.e. at the second edge of the signal sZCD (not shown in FIG. 8A) that follows the time instant $t_{transition_hi}$. The same applies in the case shown in FIG. 8B, which refers to the case $V_{out} < V_{in}/2$.

In addition, referring again, as an example, to FIG. 7, depending on the amplitude of the hysteresis, i.e. the difference $V_{out_hi} - V_{out_lo}$, the logic stage **24** may be forced to execute a certain number of pulses/semi-pulses before re-entering the pulse-skipping mode. Put in other words, referring as an example to the second semi-pulse shown in FIG. 7, the condition $v_{out} = V_{out_hi}$ cannot be met during the third phase, if the difference $V_{out_hi} - V_{out_lo}$ is too high with respect to the ripple ΔV_{pp} ; in this case, the condition $v_{out} - V_{out_hi}$ is met in a following phase.

In that concerns the resumption of the execution of the phase sequence, it is also possible that the logic stage **24** resumes the execution at a time instant different from the time instant t_{lo} , such as a time instant synchronized with an inner clock (not shown) generated by the control module **10**. As an example, the resumption of the execution of the phase sequence may occur at the first edge of the inner clock that follows the time instant t_{lo} .

It has further to be noted that the above description applies also in case the operational transconductance amplifier **12** is configured so that at steady-state the average output voltage V_{out} is greater than the output reference voltage V_{out_ref} , so that the control voltage v_c is lower than the voltage v_{c_dc} . Also, in this case, $V_{out_hi} > V_{out_lo} > V_{out_ref}$.

Referring again to FIG. 4, the control module **10** may comprise a charge-control circuit **50**, which include a first and a second charge-control transistor **52**, **54**. The first charge-control transistor **52** is a N-MOS transistor with drain set to the input voltage V_{in} and source terminal connected to the source terminal of the second charge-control transistor **54**, which is a P-MOS transistor and has a drain terminal connected to ground. The source terminals of the first and the second charge-control transistor **52**, **54** are connected to the first intermediate node N_1 .

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The logic stage **24** may activate the charge-control circuit **50** during the periods T_{skip} , by providing on the gate terminals of the first and the second charge-control transistor **52**, **54** a voltage VGH and, respectively, a voltage VGL, and by turning on the fourth transistor S_4 , whereas the first, the second and the third transistor S_1 , S_2 , S_3 are off. Therefore, by designating as V_{TH} and $|V_{TL}|$ the threshold of the first charge-control transistor **52** and, respectively, the modulus of the threshold of the second charge-control transistor **54**, the voltage V_{Cr} on the capacitor C_r is forced to stay between the voltage $V_{GH}-V_{TH}$ and the voltage $V_{GL}+|V_{TL}|$. Assuming $V_{GH}=V_{GL}=V_{in}/2$, the charge-control circuit **50** keeps the voltage V_{Cr} close to the voltage $V_{in}/2$. In facts, if the voltage V_{Cr} falls below the voltage $V_{GH}-V_{TH}$, the first charge-control transistor **52** turns on and couples the capacitor C_r to the input voltage V_{in} , while the second charge-control transistor **54** is off; on the contrary, if the voltage V_{Cr} overcomes the voltage $V_{GL}+|V_{TL}|$, the second charge-control transistor **54** turns on and couples the capacitor C_r to the ground, while the first charge-control transistor **52** is off. The voltages VGH and VGL may be obtained through a divider (not shown) connected between the input voltage V_{in} and the ground.

When the control module **10** operates in the normal mode, the logic stage **24** keeps the charge-control circuit **50** off, as an example by setting $V_{GH}=0V$ and $V_{GL}=V_{in}$.

In practice, the charge-control circuit **50** is configured to keep the voltage on the capacitor C_r close to the voltage $V_{in}/2$ also during the periods T_{skip} . In the absence of the charge-control circuit **50**, during the periods T_{skip} , the capacitor C_r could discharge; therefore, assuming as an example the case $V_{out} < V_{in}/2$, the slopes of the curves shown in FIG. 3A could differ. In particular, the slopes of the current i_L in the first and the third phase would rise, thereby causing an increase of the charge Q and thus of the ripple on the output voltage v_{out} .

In addition, the logic stage **24** may activate the charge-control circuit **50** at start-up, so as to reduce the voltage on the inductor L_r and, as a consequence, to reduce the start-up overcurrent.

In any case, the charge-control circuit **50** is optional and may be added to any of the embodiments; however, for the sake of brevity, its possible presence won't be further mentioned in this description.

From a practical point of view, as shown in FIG. 4, the control module **10** may be integrated within a single semiconductor die **29** along with the first, the second, the third and the fourth transistor S_1 , S_2 , S_3 , S_4 . The inductor L_r , the capacitor C_r and the output capacitor C_o may be arranged outside the die **29**, though embodiments are possible (not shown) in which also the inductor L_r and the capacitor C_r are integrated in the die **29**. Similarly, embodiments are possible in which the first, the second, the third and the fourth transistor S_1 , S_2 , S_3 , S_4 are arranged outside the die **29**.

FIG. 9 shows a further embodiment, hereinafter described with reference to the differences over the embodiment show in FIG. 4.

In detail, the control node N_c is set (e.g., by a bandgap circuit, not shown) to a fixed voltage V_{cfixed} , which is received by the voltage controlled delay generator **14**, along with the signals sTRIGGER and sMODE. Therefore, the signal sTIME generated by the voltage controlled delay generator **14** represents a delay T^* which is a fixed value, which depends on whether $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$. Put in other words, the durations $T_{S'}$ and T_S are fixed.

In addition, the comparator circuit (here designated by **115**) receives the output voltage v_{out} and is configured to

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compare the output voltage v_{out} with the first and the second output threshold V_{out_hi} , V_{out_lo} . In particular, the comparator circuit **115** generates a rising edge of the signal v_{skip} whenever the output voltage v_{out} overcomes the first output threshold V_{out_hi} ; in addition, the comparator circuit **115** generates a falling edge of the signal v_{skip} whenever the output voltage v_{out} drops below the second output threshold V_{out_lo} .

The functioning of the embodiment of FIG. 9 is the same as the one of FIG. 4, but for the fact that the current I_{out} (see the above equation (1)), which is a function of the fixed voltage V_{cfixed} , has to be greater than, or equal to, the current I_{LOAD} . As an example, the fixed voltage V_{cfixed} may be equal to the voltage V_{cmin} , in which case the current I_{LOAD} is equal to I_{LOAD_MIN} , which represents the maximum current that can be regulated, because the voltage controlling the voltage controlled delay generator **14** is fixed. Therefore, if $I_{LOAD} > I_{LOAD_MIN}$, the current I_{out} is still equal to the current I_{LOAD_MIN} , and the output voltage v_{out} drops to zero.

Also, the embodiment of FIG. 9 allows to implement a lower bound for durations $T_{S'}$ and T_S , as well a mechanism that provides for a reduction of the effective switching frequency, without a clamping circuit, nor a proportional integrative controller.

FIG. 10 shows a further embodiment, which is the same as the one of FIG. 4, but it does not include the selector circuit **22**. Furthermore, for the sake of simplicity, FIG. 10 does not show the operational transconductance amplifier **12**, the compensation network **15** and the (possible) charge-control circuit **50**. Furthermore, the voltage controlled delay generator **14** and the logic stage **24** are configured to operate in the same way as described above for the case sMODE='0'; put in other words, the voltage controlled delay generator **14** and the logic stage **24** are configured to operate in the scenario $V_{out} > V_{in}/2$.

This having been said, the current limiter **60** is connected in parallel to the ZCD circuit **9**. As an example, the current limiter **60** includes a first and a second operational amplifier **62**, **64**, hereinafter referred to as, respectively, the first and the second limiter amplifier **62**, **64**. Furthermore, the current limiter **60** includes a first and a second voltage generator **66**, **68**, which generate respectively a positive voltage V_{thpos} and a positive voltage V_{thneg} . Furthermore, in the following reference will be made to the third intermediate node N_3 to designate the node interposed between the inductor L_r and the capacitor C_r ; moreover, reference will be made to the fourth intermediate node N_4 to designate the node interposed between the sense resistor R_{sense} and the sense capacitor C_{sense} . The third and the fourth intermediate node N_3 , N_4 are thus connected to the negative and, respectively, the positive input terminal of the amplifier **11**.

The positive and negative terminals of the first voltage generator **66** are respectively connected to the negative input terminal of the first limiter amplifier **62** and the third intermediate node N_3 , the positive input terminal of the first limiter amplifier **62** being connected to the fourth intermediate node N_4 .

The positive and negative terminals of the second voltage generator **68** are respectively connected to the third intermediate node N_3 and the positive input terminal of the second limiter amplifier **64**, the negative input terminal of the second limiter amplifier **64** being connected to the fourth intermediate node N_4 .

Bearing in mind that, as explained before, the voltage on the sense capacitor C_{sense} has the same trend over time as the current i_L , the first and the second limiter amplifier **62**, **64** function in the same way as the amplifier **11** and generate,

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respectively, a signal sOV_{pos} and a signal sOV_{neg} . In particular, the signal sOV_{pos} is equal to '1' or '0' when the current i_L is respectively greater or lower than a current I_{max} , whose value is equal to $V_{thpos} * (R_{sense} * C_{sense}) / L_r$; the signal sOV_{neg} is equal to '1' or '0' when the current i_L is respectively lower or greater than a current I_{min} , whose value is equal to $-V_{thneg} * (R_{sense} * C_{sense}) / L_r$. To this regard, the voltages V_{thpos} and V_{thneg} may be comprised in the range 10 mV-1V.

In use the first and the second limiter amplifier **62**, **64** are configured to detect the occurrence of, respectively, a positive overcurrent and a negative overcurrent. The signals sOV_{pos} and sOV_{neg} are provided to the logic stage **24**.

In practice, though the control module **10** shown in FIG. **10** is configured to operate in the scenario $V_{out} > V_{in}/2$, it may operate also when $V_{out} < V_{in}/2$; this latter situation may temporarily occur, as an example, at the start-up.

In particular, when $V_{out} < V_{in}/2$, in normal mode the current i_L has the trend over time shown in FIG. **11**, which differs from the trend shown in FIG. **3B**. In facts, during the first phase, the current i_L rises, as opposed to decreasing, because the quantity $[(V_{in}/2) - V_{out}]/L$ is positive; furthermore, during the third phase, the current i_L decreases, as opposed to rising, because the quantity $[V_{out} - (V_{in}/2)]/L$ is negative.

In particular, the logic stage **24** is configured so that the first phase extends from the end of the preceding fourth phase till the time instant in which any of the following conditions occur: the current i_L reaches the zero value, as indicated by the signal $sZCD$, or $i_L = I_{max}$, as indicated by the signal sOV_{pos} .

The second phase extends from the end of the first phase till a time instant t_{e2} , which is delayed by a delay equal to the duration T_s from the falling edge FE-ZCD of the signal $sZCD$ that follows the end of the first phase.

The third phase extends from the end of the preceding second phase till the time instant in which any of the following conditions occur: the current i_L reaches the zero value, as indicated by the signal $sZCD$, or $i_L = I_{min}$, as indicated by the signal sOV_{neg} .

The fourth phase extends from the end of the preceding third phase till a time instant t_{e4} , which is delayed by a delay equal to the duration T_s from the rising edge RE-ZCD of the signal $sZCD$ that follows the end of the third phase.

In practice, the control module **10** makes up for the absence of the selector circuit **22** by limiting the current i_L , when $V_{out} < V_{in}/2$, in order to prevent the voltage on the capacitor C_r and the current i_L from rising too much, in which case it would take much time for the current i_L to get back to zero, thereby causing a long charge of the capacitor C_r .

Furthermore, referring to the situation shown in FIG. **11**, also in this case the execution of the phase sequence can be suspended when the current i_L is null. As an example, in case the signal v_{skip} goes to '1' in a time instant $t_{transition_hi}$ that falls between the rising edge RE-ZCD and the falling edge FE-ZCD of the signal $sZCD$ (i.e., during the first phase or a first part of the fourth phase, with positive current i_L , or a first part of the second phase, with positive current i_L), the phase sequence can be suspended at the following falling edge FE-ZCD of the signal $sZCD$, which falls within the second phase, i.e. after the execution of the abovementioned first part of the second phase; furthermore, the phase sequence will be resumed by carrying out a second part of the second phase, with duration equal to the duration T_s and negative current i_L . In case (not shown) the signal v_{skip} goes to '1' in a time instant that falls during the second part of the

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second phase or the third phase or a second part of the fourth phase, this latter featuring negative current i_L , the phase sequence can be suspended at the following rising edge RE-ZCD of the signal $sZCD$, which falls within the fourth phase, i.e. after the execution of the second part of the fourth phase; furthermore, the phase sequence will be resumed by carrying out the first part of the fourth phase, with duration equal to the duration T_s and positive current i_L .

FIG. **12** shows a further embodiment, which differs from the one of FIG. **4** in that the clamping circuit **22**, the operational transconductance amplifier **12** and the compensation network **15** are absent. Furthermore, the control module (here designated by **210**) includes an analog-to-digital converter (ADC) **200**, a proportional integrative controller **212**, a digital-to-time converter **214** and the comparator circuit **115**.

The input of the analog-to-digital converter **200** is connected to the output node N_{OUT} , in order to receive the output voltage v_{out} . The analog-to-digital converter **200** converts the output voltage v_{out} into a digital signal $sNUM_{out}$, which represents a corresponding numerical value NUM_{out} .

The proportional integrative controller **212** receives the signal $sNUM_{out}$, a digital signal $sNUM_{ref}$ and a clock signal CLK; as an example, the digital signal $sNUM_{ref}$ and the clock signal CLK may be generated, respectively, by a reference generator and a timing circuit of the control module **210**, not shown.

The signal $sNUM_{ref}$ represents a corresponding numerical value NUM_{ref} . The numerical value NUM_{ref} may be equal, as an example, to the numerical value that would be generated by the analog-to-digital converter **200** if this latter received at input the output reference voltage V_{out_ref} .

The clock CLK provides the timing to the proportional integrative controller **212**, which generates a control signal $sNUM_c$ of the digital type, which represents a corresponding numerical value NUM_c and is provided to the digital-to-time converter **214** along with the abovementioned signals **STRIGGER** and **sMODE**.

In detail, by referring to NUM_{err} to designate $NUM_{ref} - NUM_{out}$ and by adopting the symbol 'j' to index the samples of NUM_{err} and NUM_c , $NUM_c(j)$ is set equal to the greater of a threshold value NUM_{cmin} and:

$$K_{prop} * NUM_{err}(j) + K_{int} * \text{sum}_{u=0}^j [NUM_{err}(u)] \quad (8)$$

wherein: "sum" represents the sum of the corresponding addends, which are indexed by the index $u=0, 1, \dots, j$; K_{prop} and K_{int} are positive constants; $NUM_{err}(j) = NUM_{ref} - NUM_{out}(j)$.

As an example, the sampling period of the signal $sNUM_{out}$ and of the control signal $sNUM_c$ may be equal to the period of the clock signal CLK. Furthermore, the equation (8) may be modified to limit the number of previous samples which are considered to compute the third addend.

In practice, the proportional integrative controller **212** implements a numerical limiting mechanism equivalent to the analog one of the embodiments of FIG. **4**, this latter being based on the voltage V_{cmin} .

The digital-to-time converter **214** is configured to generate the signal $sTIME$ so that the delay T^* represented by the signal $sTIME$ is equal $k_{DTC} * NUM_c(j)$, wherein k_{DTC} is a constant which is a function of the period of the clock signal CLK and depends on the value of the signal **sMODE**. As an example, k_{DTC} may be equal to $1/(MULT * T_{clock})$, with T_{clock} representing the period of the clock signal CLK and **MULT** assuming a first or a second predetermined value, respectively when **sMODE** is equal to '1' or '0'.

Basically, the digital-to-time converter **214** converts a number into a delay; to this end, it may be formed by a respective counter which is updated with a high-frequency clock (i.e., with a clock having a frequency higher than the frequency of the clock signal CLK), the counter being compared with a corresponding threshold which depends on $NUM_c(j)$. As an example, at each rising edge of the signal sTRIGGER, a falling edge of the signal sTIME is generated and the counter is reset; a corresponding rising edge of the signal sTIME is then generated when the counter reaches a threshold that is equal to $NUM_c(j)$. In any case, the technical implementation of the digital-to-time converter **214** is irrelevant for the implementation of various embodiments.

The signal sTIME is provided to the logic stage **24**, along with the signal v_{skip} , which is generated by the comparator circuit **115** in the same way described with reference to FIG. **9**, namely by comparing the output voltage v_{out} with the first and the second output threshold V_{out_hi} , V_{out_lo} .

In this embodiment, I_{LOAD_MIN} represents the value of the current I_{LOAD} at steady-state such that $NUM_c(j)=NUM_{cmin}$. Therefore, in use, if the current I_{LOAD} drops below the value I_{LOAD_MIN} , the proportional integrative controller **212** sets $NUM_c(j)=NUM_{cmin}$. As a consequence, the current I_{out} becomes greater than the current I_{LOAD} , and the output voltage v_{out} rises as discussed with reference to FIGS. **6** and **7**. NUM_{out} replicates the output voltage v_{out} , so, when $I_{LOAD} > I_{LOAD_MIN}$, NUM_{out} is approximately equal to NUM_{ref} ; when $I_{LOAD} < I_{LOAD_MIN}$, NUM_{out} rises in the same way as the output voltage v_{out} .

The logic stage **24** operates in the same manner described with reference to the previous embodiments. Therefore, the embodiment shown in FIG. **12** represents a partially numerical implementation of the embodiment shown in FIG. **4**. The same considerations on possible variations of the timing of the stop/resumption of the execution of the phase sequence apply. Furthermore, embodiments are possible which are the same as the one shown in FIG. **12**, but do not include the selector circuit **22**, in which case the current limiter **60** may be included.

Finally, it is clear that, as mentioned previously, modifications and variations may be made to what has been described and illustrated herein, without departing from the scope of the present invention, as defined in the annexed claims.

For instance, the switches of the converter may be formed by transistors of a different type with respect to the type of the transistors mentioned before. In addition, the first, the second, the third and the fourth transistor S_1 , S_2 , S_3 , S_4 may be formed by different types of transistors; as an example, embodiments are possible in which the first and the third transistor S_1 , S_3 are P-MOS transistors, whereas the second and the fourth transistor S_2 , S_4 are N-MOS transistors.

The compensation network **15** may include further electrical components, such as an additional capacitor connected in parallel to the series circuit formed by the resistor R_c and the capacitor C_c , in order to reduce the loop gain at high frequencies and improve stability.

In addition, the compensation network may be such that the relationship between the control voltage v_c and the error voltage v_{err} is not of the proportional and integrative type. The same applies in case of the above equation (8). From a more general standpoint, by referring to the control quantity to designate either i) the control voltage v_c or ii) the numerical value NUM_c , and referring to the error quantity to designate either i) the error voltage v_{err} or ii) NUM_{err} , the dependence of the control quantity on the error quantity may be different from the proportional integrative dependence.

Similarly, the dependence of the delay T^* on the control voltage v_c and, respectively, the numerical value NUM_c may be not linear, but generically monotonically increasing.

Instead of providing for a voltage controlled delay generator capable of generating, given any value of the control voltage v_c , different values of the delay T^* for the scenarios $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$, it is possible to adapt the output dynamic range of the operational transconductance amplifier **12**, based on whether $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$, in order to generate, given any value of the error voltage v_{err} , a value of the control voltage v_c which depends on whether $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$; in this case, the voltage controlled delay generator may not have to operate in different manner based on whether $V_{out} < V_{in}/2$ or $V_{out} > V_{in}/2$.

Finally, instead of using the output voltage v_{out} as a feedback quantity to generate the error quantity, it is possible to use feedback quantity v_{out_fb} , which depends on the output voltage v_{out} in a monotonic way, either increasing or decreasing; in particular, in case of decreasing dependence, the inputs of the operational transconductance amplifier **12** have to be inverted, to maintain a negative feedback. As an example, the feedback quantity v_{out_fb} may be obtained as a partition of the output voltage v_{out} as an example by means of a resistive divider. The feedback quantity v_{out_fb} is fed to the operational transconductance amplifier **12** or the analog-to-digital converter **200**.

Finally, it is possible for the dependence of the time delay T^* to be monotonically decreasing, instead of monotonically increasing. In this case, referring as an example to the control module **10** shown in FIG. **4**, the clamping circuit may be modified so as to be activated in case the control voltage v_c tends to overcome a voltage V_{cmax} ; furthermore, the output voltage v_{out} and the output reference voltage V_{out_ref} are respectively provided on the positive and negative input terminals of the operational transconductance amplifier **12**. The current I_{LOAD_MIN} represents thus the value of the current I_{LOAD} such that the control voltage v_c is equal to the voltage V_{cmax} ; if $I_{LOAD} > I_{LOAD_MIN}$, the control voltage v_c is lower than the voltage V_{cmax} and such that the corresponding value of the delay T^* causes the current I_{out} to be equal to the current I_{LOAD} . If $I_{LOAD} < I_{LOAD_MIN}$, the control voltage v_c is clamped to the voltage V_{cmax} by the clamping circuit, so that the corresponding value of the delay T^* causes the current I_{out} to be greater than the current I_{LOAD} , and the output voltage v_{out} rises as discussed with reference to FIGS. **6** and **7**. Furthermore, the current i_{clamp} has an opposite direction with respect to the one described before (i.e., it is drained by the clamping circuit from the control node N_c instead of being injected), because it has to offset the operational transconductance amplifier **12**, which tends to raise the control voltage v_c .

The same considerations apply, as an example, in case of the embodiment of FIG. **12**: the dependence of the delay T^* on the numerical value NUM_c may be monotonically decreasing, instead of monotonically increasing, in which case the proportional integrative controller **212** may be modified so that $NUM_c(j)$ is set equal to the lower of a threshold value NUM_{cmax} and the result of the above equation (8); furthermore, NUM_{err} has to be modified into $NUM_{out} - NUM_{ref}$. I_{LOAD_MIN} represents the value of the current I_{LOAD} at steady-state such that $NUM_c(j)=NUM_{cmax}$. Therefore, in use, if $I_{LOAD} > I_{LOAD_MIN}$, $NUM_c(j)$ is lower than NUM_{cmax} and such that the corresponding value of the delay T^* causes the current I_{out} to be equal to the current I_{LOAD} ; if the current I_{LOAD} drops below the value I_{LOAD_MIN} , the proportional integrative controller **212** clamps $NUM_c(j)$ to NUM_{cmax} . As a consequence, the current I_{out} becomes

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greater than the current I_{LOAD} , and the output voltage v_{out} rises as discussed with reference to FIGS. 6 and 7.

What is claimed is:

1. A control module, for a resonant switched-capacitor converter comprising first, second, third, and fourth switches connected in cascade, wherein the first and second switches form a first inner node, the second and third switches form an output node configured to be coupled to a load and an output capacitor, and the third and fourth switches form a second inner node, the converter configured to receive an input voltage and generate an output voltage on the output node, and comprising a resonant series circuit, which is coupled to the first and second inner nodes and includes an inductor and a first capacitor, the control module comprising:

a controller stage configured to control the first, second, third, and fourth switches so that the converter operates alternatively in:

a first phase, in which the first and third switches are closed, and the second and fourth switches are open, so that an inductor current that flows through the inductor is coupled to the load;

a second phase, in which the second and third switches are closed, and the first and fourth switches are open, so that the inductor current is decoupled from the load;

a third phase, in which the second and fourth switches are closed, and the first and third switches are open, so that the inductor current is coupled to the load; and

a fourth phase, in which the first and fourth switches are closed, and the second and third switches are open, so that the inductor current is decoupled from the load;

an input stage configured to generate a first control signal indicating a control quantity;

a delay stage configured to receive the first control signal and to generate a duration signal indicating a time quantity that depends on the control quantity according to a first function that is monotonically increasing or decreasing;

a zero-crossing detection circuit coupled to the inductor and configured to generate a current signal indicating zero crossings of the inductor current;

wherein the control quantity is fixed, or variable and depends on a difference between a reference quantity and a feedback quantity that depends on the output voltage;

wherein the input stage is further configured, when the control quantity is variable, to clamp the control quantity to a control threshold in response to the control quantity reaching the control threshold, wherein the control threshold is a lower bound for the control quantity when the first function is monotonically increasing, or is an upper bound for the control quantity when the first function is monotonically decreasing;

wherein the controller stage is further configured to:

receive the current signal and the duration signal; and operate in a normal mode, in which the controller stage controls the first, second, third, and fourth switches to carry out a phase sequence by iterating pulses, each respective pulse including a respective first phase, a respective second phase, a respective third phase, and a respective fourth phase, the respective first, second, third and fourth phases having respective timings that depend on the zero crossings of the inductor current and on the time quantity, so that a

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duration of each pulse rises as the time quantity rises and the converter generates on the output node an average output current directed to the load and the output capacitor that depends on the time quantity according to a monotonically increasing second function, wherein the average output current is prevented from dropping below a minimum load current, which is a function of the control quantity when the control quantity is fixed, and of the control threshold, when the control quantity is variable, wherein the output voltage thereby tends to rise when the load drains a load current that is lower than the minimum load current;

a comparison stage configured to generate a second control signal indicating time instants in which the output voltage reaches an upper voltage threshold and time instants in which the output voltage drops to a lower voltage threshold; and

wherein the controller stage is further configured to:

receive the second control signal;

when the second control signal indicates that the output voltage has reached the upper voltage threshold, to switch from the normal mode to a pulse-skipping mode, in which the controller stage suspends execution of the phase sequence, thereby causing the output voltage to drop; and

resume the execution of the phase sequence after the second control signal indicates that the output voltage has dropped to the lower voltage threshold.

2. The control module according to claim 1, wherein, after each time instant in which the output voltage reaches the upper voltage threshold, the controller stage is configured to suspend the execution of the phase sequence at a corresponding time instant that follows the time instant and in which the inductor current is null.

3. The control module according to claim 2, wherein the controller stage is configured to suspend the execution of the phase sequence at a first time instant that follows the time instant in which the output voltage reaches the upper voltage threshold and in which the inductor current is null.

4. The control module according to claim 1, wherein, when operating in the normal mode, the controller stage is configured to control the first, second, third, and fourth switches so as to carry out at least one of first and second phase sequences, the first phase sequence being formed by an iteration of first pulses, each first pulse being formed by a succession of the first, second, third, and fourth phases, the second phase sequence being formed by an iteration of second pulses, each second pulse being formed by a succession of the fourth, first, second, and third phases; the controller stage being further configured to set durations of the first, second, third and fourth phases so that:

when the controller stage carries out the first phase sequence, each of the first and third phases starts when the inductor current is null and has a duration that is equal to the time quantity, each second phase lasting from an end of a previous first phase until when the inductor current goes to zero, each fourth phase lasting from an end of a previous third phase until when the inductor current goes to zero; and

when the controller stage carries out the second phase sequence, the duration of the second and fourth phases is equal to the time quantity, each first phase lasting from an end of a previous fourth phase until when the inductor current goes to zero, each third phase lasting from an end of a previous second phase until when the inductor current goes to zero.

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5. The control module according to claim 4, further comprising a selector circuit configured to detect whether an average of the output voltage is greater or lower than a discrimination voltage proportional to the input voltage and to control the controller stage so that the controller stage carries out either the first or second phase sequence, as a function of an outcome of the detection.

6. The control module according to claim 1, wherein, when operating in the normal mode, the controller stage is configured to control the first, second, third, and fourth switches so as to carry out a third phase sequence formed by an iteration of pulses, each pulse being formed by a succession of the fourth, first, second, and third phases; the controller stage being further configured to set durations of the first, second, third and fourth phases so that:

each first phase lasts from an end of a previous fourth phase until when alternatively the inductor current goes to zero or the inductor current reaches an upper current threshold;

each second phase starts at an end of a previous first phase and, alternatively, has a duration equal to the time quantity, if the previous first phase has ended with a null inductor current, or ends at a time instant delayed by a delay equal to the time quantity with respect to a first zero crossing of the inductor current following the end of the previous first phase, if the previous first phase has ended with inductor current equal to the upper current threshold;

each third phase lasts from an end of a previous second phase until when alternatively the inductor current goes to zero or the inductor current drops to a lower current threshold; and

each fourth phase starts at an end of a previous third phase and, alternatively, has the duration equal to the time quantity, if the previous third phase has ended with the null inductor current, or ends at a time instant delayed by a delay equal to the time quantity with respect to the first zero crossing of the inductor current following the end of the previous third phase, if the previous third phase has ended with inductor current equal to the lower current threshold.

7. The control module according to claim 1, further comprising a charge-control circuit configured to keep a voltage on the first capacitor of the resonant series circuit within a range including half the input voltage.

8. The control module according to claim 1, wherein the input stage comprises a transconductance amplifier and a compensation network, the transconductance amplifier being configured to generate a control voltage on the compensation network, as a function of the difference between a reference voltage and a feedback voltage that monotonically depends on the output voltage, the control quantity being variable and being formed by the control voltage.

9. The control module according to claim 1, wherein the input stage comprises:

an analog-to-digital converter configured to generate a digital signal representing an output numerical value, which is a function of output voltage; and

a numerical controller configured to receive the digital signal and to generate the first control signal so that it represents a numerical control value, which is a function of the difference between the output numerical value and a numerical reference, the control quantity being variable and being formed by the numerical control value.

10. The control module according to claim 1, wherein the comparison stage is configured to generate the second

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control signal so that it indicates the time instants in which a monitoring quantity reaches a respective first control threshold and the time instants in which the monitoring quantity reaches a respective second control threshold, the monitoring quantity depending monotonically on the output voltage.

11. A system comprising:

a resonant switched-capacitor converter comprising:

first, second, third, and fourth switches connected in cascade, wherein the first and second switches form a first inner node, the second and third switches form an output node configured to be coupled to a load and an output capacitor, and the third and fourth switches form a second inner node, the converter configured to receive an input voltage and generate an output voltage on the output node; and

a resonant series circuit, which is coupled to the first and second inner nodes and includes an inductor and a first capacitor; and

a control module comprising:

a controller stage configured to control the first, second, third, and fourth switches so that the converter operates alternatively in:

a first phase, in which the first and third switches are closed, and the second and fourth switches are open, so that an inductor current that flows through the inductor is coupled to the load;

a second phase, in which the second and third switches are closed, and the first and fourth switches are open, so that the inductor current is decoupled from the load;

a third phase, in which the second and fourth switches are closed, and the first and third switches are open, so that the inductor current is coupled to the load; and

a fourth phase, in which the first and fourth switches are closed, and the second and third switches are open, so that the inductor current is decoupled from the load;

an input stage configured to generate a first control signal indicating a control quantity;

a delay stage configured to receive the first control signal and to generate a duration signal indicating a time quantity that depends on the control quantity according to a first function that is monotonically increasing or decreasing;

a zero-crossing detection circuit coupled to the inductor and configured to generate a current signal indicating zero crossings of the inductor current;

wherein the control quantity is fixed, or variable and depends on a difference between a reference quantity and a feedback quantity that depends on the output voltage;

wherein the input stage is further configured, when the control quantity is variable, to clamp the control quantity to a control threshold in response to the control quantity reaching the control threshold, wherein the control threshold is a lower bound for the control quantity when the first function is monotonically increasing, or is an upper bound for the control quantity when the first function is monotonically decreasing;

wherein the controller stage is further configured to: receive the current signal and the duration signal; and

operate in a normal mode, in which the controller stage controls the first, second, third, and fourth

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switches to carry out a phase sequence by iterating pulses, each respective pulse including a respective first phase, a respective second phase, a respective third phase, and a respective fourth phase, the respective first, second, third and fourth phases having respective timings that depend on the zero crossings of the inductor current and on the time quantity, so that a duration of each pulse rises as the time quantity rises and the converter generates on the output node an average output current directed to the load and the output capacitor that depends on the time quantity according to a monotonically increasing second function, wherein the average output current is prevented from dropping below a minimum load current, which is a function of the control quantity when the control quantity is fixed, and of the control threshold, when the control quantity is variable, wherein the output voltage thereby tends to rise when the load drains a load current that is lower than the minimum load current;

a comparison stage configured to generate a second control signal indicating first time instants in which the output voltage reaches an upper voltage threshold and second time instants in which the output voltage drops to a lower voltage threshold; and wherein the controller stage is further configured to: receive the second control signal;

when the second control signal indicates that the output voltage has reached the upper voltage threshold, to switch from the normal mode to a pulse-skipping mode, in which the controller stage suspends execution of the phase sequence, thereby causing the output voltage to drop; and

resume the execution of the phase sequence after the second control signal indicates that the output voltage has dropped to the lower voltage threshold.

12. The system according to claim 11, further comprising a charge-control circuit configured to keep a voltage on the first capacitor of the resonant series circuit within a range including half the input voltage.

13. The system according to claim 11, wherein the input stage comprises a transconductance amplifier and a compensation network, the transconductance amplifier being configured to generate a control voltage on the compensation network, as a function of the difference between a reference voltage and a feedback voltage that monotonically depends on the output voltage, the control quantity being variable and being formed by the control voltage.

14. The system according to claim 11, wherein the input stage comprises:

an analog-to-digital converter configured to generate a digital signal representing an output numerical value, which is a function of output voltage; and

a numerical controller configured to receive the digital signal and to generate the first control signal so that it represents a numerical control value, which is a function of the difference between the output numerical value and a numerical reference, the control quantity being variable and being formed by the numerical control value.

15. A method for controlling a resonant switched-capacitor converter comprising a first, a second, a third and a fourth switch connected in cascade, so that the first and second switches form a first inner node, the second and third switches form an output node configured to be coupled to a

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load and an output capacitor, and the third and fourth switches form a second inner node, the converter being configured to receive an input voltage and generate an output voltage on the output node and comprising a resonant series circuit, which is coupled to the first and second inner nodes and includes an inductor and a first capacitor, the method comprising:

controlling the first, second, third, and fourth switches so that the converter operates alternatively in:

a first phase, in which the first and third switches are closed, and the second and fourth switches are open, so that an inductor current that flows through the inductor is coupled to the load;

a second phase, in which the second and third switches are closed, and the first and fourth switches are open, so that the inductor current is decoupled from the load;

a third phase, in which the second and fourth switches are closed, and the first and third switches are open, so that the inductor current is coupled to the load; and

a fourth phase, in which the first and fourth switches are closed, and the second and third switches are open, so that the inductor current is decoupled from the load;

generating a first control signal indicating a control quantity; and

based on the first control signal, generating a duration signal indicating a time quantity that depends on the control quantity according to a first function that is monotonically increasing or decreasing; and

generating a current signal indicating zero crossings of the inductor current;

the control quantity being fixed, or variable and depending on a difference between a reference quantity and a feedback quantity that depends on the output voltage;

in response to the control quantity being variable, clamping the control quantity to a control threshold, in response to the control quantity reaching the control threshold, the control threshold being a lower bound for the control quantity, in response to the first function is monotonically increasing, or an upper bound for the control quantity, in response to the first function is monotonically decreasing; the method further comprising operating the converter in a normal mode, based on the current signal and the duration signal, the step of operating the converter in the normal mode comprising controlling the first, second, third, and fourth switches so as to carry out a phase sequence by iterating pulses including, each, a respective first phase, a respective second phase, a respective third phase and a respective fourth phase, the respective first, second, third and fourth phases having respective timings which depend on the zero crossings of the inductor current and on the time quantity, so that a duration of each pulse rises as the time quantity rises and the converter generates on the output node an average output current directed to the load and the output capacitor that depends on the time quantity according to a monotonically increasing second function, the average output current being prevented from dropping below a minimum load current, which is a function of the control quantity, in response to the control quantity being fixed, and of the control threshold, in response to the control quantity is variable, the output voltage thereby tending to rise in case the load drains a load current that is lower than the minimum load current; the method further comprising:

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generating a second control signal indicating time instants in which the output voltage reaches an upper voltage threshold and time instants in which the output voltage drops to a lower voltage threshold; and

in response to the second control signal indicating that the output voltage has reached the upper voltage threshold, switching from operating the converter in the normal mode to operating the converter in a pulse-skipping mode, the operating the converter in the pulse-skipping mode comprising suspending execution of the phase sequence, thereby causing the output voltage to drop; and

resuming the execution of the phase sequence after the second control signal indicates that the output voltage has dropped to the lower voltage threshold.

16. The method according to claim **15**, wherein, after each time instant in which the output voltage reaches the upper voltage threshold, execution of the phase sequence is suspended at a corresponding time instant that follows the time instant and in which the inductor current is null.

17. The method according to claim **16**, wherein the execution of the phase sequence is suspended at a first time instant that follows the time instant in which the output voltage reaches the upper voltage threshold and in which the inductor current is null.

18. The method according to claim **15**, wherein operating the converter in the normal mode comprises controlling the first, second, third, and fourth switches so as to carry out at least one of first and second phase sequences, the first phase sequence being formed by an iteration of first pulses, each first pulse being formed by a succession of the first, second, third, and fourth phases, the second phase sequence being formed by an iteration of second pulses, each second pulse being formed by a succession of the fourth, first, second, and third phases; the method comprising setting durations of the first, second, third, and fourth phases so that:

in response to the first phase sequence being carried out, each of the first and third phases starts in response to the inductor current being null and has a duration that is equal to the time quantity, each second phase lasting from an end of a previous first phase until the inductor current goes to zero, each fourth phase lasting from an end of a previous third phase until the inductor current goes to zero; and

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in response to the second phase sequence being carried out, the duration of the second and fourth phases is equal to the time quantity, each first phase lasting from an end of a previous fourth phase until the inductor current goes to zero, each third phase lasting from an end of a previous second phase until the inductor current goes to zero.

19. The method according to claim **18**, further comprising detecting whether an average of the output voltage is greater or lower than a discrimination voltage proportional to the input voltage and carrying out either the first or second phase sequence, as a function of an outcome of the detection.

20. The method according to claim **15**, wherein operating the converter in the normal mode comprises controlling the first, second, third, and fourth switches so as to carry out a third phase sequence formed by an iteration of pulses, each pulse being formed by a succession of the fourth, first, second, and third phases; the method comprising setting durations of the first, second, third and fourth phases so that:

each first phase lasts from an end of a previous fourth phase until the inductor current goes to zero or the inductor current reaches an upper current threshold;

each second phase starts at an end of a previous first phase and, alternatively, has a duration equal to the time quantity, in response to the previous first phase having ended with a null inductor current, or ends at a time instant delayed by a delay equal to the time quantity with respect to a first zero crossing of the inductor current following the end of the previous first phase, in response to the previous first phase having ended with inductor current equal to the upper current threshold;

each third phase lasts from an end of a previous second phase until the inductor current goes to zero or the inductor current drops to a lower current threshold; and

each fourth phase starts at an end of a previous third phase and, alternatively, has the duration equal to the time quantity, in response to the previous third phase having ended with the null inductor current, or ends at a time instant delayed by a delay equal to the time quantity with respect to the first zero crossing of the inductor current following the end of the previous third phase, in response to the previous third phase having ended with inductor current equal to the lower current threshold.

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